Flash Flood Susceptibility Mapping in Nairobi Watershed: A Comparison of Shannon's Entropy and Frequency Ratio Models

A RESESERCH COLLABORATION

By

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Thankyou. Sincerely,

Mutten Joan & Maxwell Lemeriwas

4 Abstract

Flooding in Nairobi, Kenya, presents significant socio-economic and environmental challenges, exacerbated by climate change and rapid urbanization. This study focuses on predicting flood susceptibility using two statistical models: Shannon's Entropy (SE) and Frequency Ratio (FR). Both models integrate diverse flood conditioning factors, including elevation, slope, rainfall, land use, drainage, and river density, to identify high-risk areas. Sentinel-1 and Sentinel-2 imagery, combined with data from CHIRPS and SRTM DEM, facilitated precise spatial analysis. The results reveal distinct mapping characteristics, with the SE model achieving higher predictive accuracy, reflected by an AUC score of 0.930, compared to 0.832 for the FR model. Additionally, a Google Earth Engine (GEE)-based predictive tool was developed, leveraging dynamic rainfall data to assess flood risks in real time, delineate flood-prone areas, and estimate affected populations. These findings underscore the importance of robust flood susceptibility mapping and predictive tools for effective urban planning and disaster management. By identifying high-risk zones, this research aims to guide policy decisions to enhance Nairobi's resilience to flooding and ensure sustainable urban development.

5 TABLE OF CONTENTS

4 Contents

3 Acknowledgements iii

4 Abstract iv

5 TABLE OF CONTENTS v

5 List of tables viii

5 List of tables ix

Acronyms and abbreviations x

7 1 INTRODUCTION 1

1.1 Background Study 1

1.2 Problem Statement 2

1.3 Research Identification 2

1.4 Research Objectives 3

1.5 Justification of the Study 3

8 2 LITERATURE REVIEW 5

2.1 Introduction 5

2.2 Remote Sensing and Geospatial Technologies for Flash Flood Analysis 7

2.2.1 Digital Elevation Models (DEMs) 7

2.2.2 Synthetic Aperture Radar (SAR) 7

2.2.3 Optical Remote Sensing 8

2.2.4 Google Earth Engine (GEE) 8

2.2.5 Geographic Information Systems (GIS) 8

2.2.6 R 8

9 3 MATERIALS AND METHODS 9

3.1 Study Area 9

3.3 Methodology 13

3.3.1 Flood inventory map using GEE 14
3.3.2 Flood Inventory processing in R 17
3.4 Flood conditioning factors 19
3.4.1 Slope Angle 19
3.4.2 Slope Aspect 20
3.4.3 Plan Curvature 21
3.4.4 Altitude 22
3.4.5 River Density 23
3.4.6 Drainage Density 24
3.4.7 Distance from Rivers 25
3.4.8 Road Density 26
3.4.9 LULC 27
3.4.10 NDVI 28
3.4.11 Rainfall 29
3.4.12 Soil 30
3.5.1 Correlation matrix 31
3.5.2 Multicollinearity analysis and explanatory factor selection 33
3.6 Flood susceptibility Mapping 33
3.6.1 FR Model 33
3.6.2 SE Model 34
10 4 RESULTS ..... - 43 -
4.1 Flood Susceptibility Maps ..... - 43 -
4.2 Ranked Factors Influencing Flood Susceptibility in Nairobi ..... - 44 -
4.3 Training and testing distribution ..... - 47 -
4.4 Model Validation and Evaluation ..... - 49 -
4.5 Model Validation Using the AUC-ROC Curve Method using testing data ..... - 51 -

4.6 Flood prediction simulation using GEE ..... - 55 -

11 5.DISCUSSION ..... - 57 -

12 6 CONCLUSION AND RECOMENDATION ..... - 59 -

13 7 REFERENCES ..... - 60 -

14 8 APPENDIXES ..... - 63 -

5 List of tables

Figure 1:Image by BBC 4
Figure 2:Image by Climate Change Writers 4
Figure 3:Study Area 10
Figure 4:Methodology flowchart 13
Figure 5:Flood Map GEE 16
Figure 6:Flood distribution and river density 18
Figure 7:slope map 19
Figure 8:Aspect map 20
Figure 9:Curvature map 21
Figure 10:Elevation map 22
Figure 11:River Density 23
Figure 12:Drainage density 24
Figure 13:Distance from rivers 25
Figure 14:Road Density 26
Figure 15:Land Use Land Cover 27
Figure 16:NDVI 28
Figure 17:Rainfall above 80mm 29
Figure 18:Soil types 30
Figure 19:Correlation heat map 31
Figure 20:Flood susceptibility map using FR ..... - 43 -
Figure 21:Flood susceptibility map using SE ..... - 44 -
Figure 22:Training,validation and area coverage FR model ..... - 48 -
Figure 23:Training, validation and area coverage SE model ..... - 49 -
Figure 24:ROC curve using primary data FR model ..... - 51 -
Figure 25:ROC curve using primary data SE model ..... - 52 -
Figure 26:ROC curve using secondary data both model ..... - 54 -
Figure 27:Flood Simulation of M-A-M period ..... - 56 -
Figure 28:Flood simulation January 2025Figure 29:Flood Simulation of M-A-M period ..... - 56 -
Figure 30:Flood simulation January 2025 ..... - 57 -
Figure 31:Flood simulation January 2025 ..... - 57 -

5 List of tables

Table 1:Data.....11
Table 2:Sentinel-1 SAR Data Acquisition Details for Flood Analysis: Pre- and Post-Flood Observations ...16
Table 3:Flood Classification Results: Pixel Count and Area of Flooded and Non-Flooded Regions 18
Table 5:Variance Inflation Factor (VIF) Values for Flood Susceptibility Model Variables 33
Table 6:Slope Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 36 -
Table 7:Aspect Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 36 -
Table 8:Curvature Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 37 -
Table 9:Elevation Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 37 -
Table 10:LULC Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 38 -
Table 11:NDVI Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 38 -
Table 12:Soil Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 39 -
Table 13:River Density Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 40 -
Table 14:Road Density Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 41 -
Table 15:Drainage Density Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 41 -
Table 16:Distance to rivers Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 42 -
Table 17:Rainfall Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters ..... - 42 -

Acronyms and abbreviations

AUC: Area Under the Curve

CHIRPS: Climate Hazards Group InfraRed Precipitation with Station data

DEM: Digital Elevation Model

FR: Frequency Ratio

FSM: Flood Susceptibility Map

GEE: Google Earth Engine

GIS: Geographic Information Systems

LULC: Land Use and Land Cover

MNDWI: Modified Normalized Difference Water Index

NDVI: Normalized Difference Vegetation Index

ROC: Receiver Operating Characteristic

SAR: Synthetic Aperture Radar

SE: Shannon's Entropy

SRTM: Shuttle Radar Topography Mission

VIF: Variance Inflation Factor

FSI: Flood Susceptibility Index

FSI_FR: Flood Susceptibility Index using Frequency Ratio

FSI_SE: Flood Susceptibility Index using Shannon's Entropy

Dist_Riv: Distance to River

7 1 INTRODUCTION

1.1 Background Study

Floods represent a critical natural hazard globally, with the potential to cause immense destruction to lives, property, and the environment. In particular, flash floods are characterized by their sudden onset and severe impacts, often exacerbated by climate change and rapid urbanization. Flash floods typically occur within six hours of heavy rainfall or other hydrological events, leading to swift water level rises and fast-moving water that can overwhelm urban infrastructure. In Nairobi, these events are most common during the long rains from March to May and the short rains from October to December. As urban areas expand, especially in regions like Nairobi, Kenya, the need for accurate flood susceptibility mapping becomes paramount to mitigate potential disasters effectively.

Nairobi, the capital city of Kenya, faces increasing flood risks due to its rapid urban development and changes in land use. The city has experienced significant flooding, with torrential rains and flash floods causing deadly havoc across the country. For instance, as of May 3, 2024, approximately 210 people died, 164 injured, 72 missing, and nearly 205,000 people were affected by heavy rains and flooding, according to the National Disaster Operations Centre (NDOC). Additionally, 194,305 people have been displaced, and extensive damage has been reported to livestock, croplands, roads, small businesses, schools, and health facilities. In Nairobi County alone, an estimated 147,061 people have been affected, and 20,968 families displaced by the heavy rainfall.

Identifying flood-prone areas within Nairobi is crucial for urban planning and disaster management. This study aims to compare two prominent statistical models, Shannon's Entropy (SE) and Frequency Ratio (FR) to assess their effectiveness in predicting flood susceptibility in Nairobi.

Shannon's Entropy (SE) is a widely used bivariate statistical method in flood susceptibility modeling, known for its ability to quantify the uncertainty and spatial distribution of flood-related factors. The method has been successfully applied in various studies, demonstrating its robustness in handling complex environmental data. For instance, Khosravi et al. (2016) employed SE alongside other models to predict flash flood susceptibility in Iran, highlighting its accuracy in identifying flood-prone areas.

Frequency Ratio (FR), another bivariate statistical approach, evaluates the spatial correlation between flood occurrences and conditioning factors. By calculating the ratio of the probability of occurrence to non-occurrence for each factor, FR provides a straightforward and interpretable means of flood susceptibility

mapping. Previous studies, such as those by Pradhan et al. (2013), have shown FR's efficacy in flood prediction, especially when combined with other statistical methods.

1.2 Problem Statement

Flash floods in Nairobi have severe consequences, including loss of life, property damage, displacement of populations, and disruptions to essential services. The most recent data indicates a high number of fatalities, injuries, and significant displacement of residents. Additionally, the economic impact is profound, and damage to infrastructure such as roads, businesses, schools, and health facilities. This situation underscores the urgent need for effective flood management and mitigation strategies.

Despite various efforts, current flood management practices in Nairobi remain largely reactive, focusing on emergency response and recovery rather than proactive measures to prevent and mitigate flood risks. This reactive approach is insufficient, given the increasing frequency and intensity of flood events attributed to climate change and urban expansion. Therefore, there is a pressing need for comprehensive flood susceptibility mapping to identify high-risk areas and inform targeted interventions.

This study aims to compare SE and FR models to determine which provides more accurate and reliable predictions of flood-prone areas in Nairobi. The findings will inform the development of more accurate flood risk assessments and enhance the city's capacity to manage and mitigate flood impacts. By identifying high-risk areas, the study will guide urban planners and policymakers in implementing effective flood mitigation strategies, such as improving drainage infrastructure, regulating land use, and developing early warning systems. This proactive approach will help reduce the vulnerability of Nairobi's residents and infrastructure to flooding, promote a safer and more resilient urban environment, and ensure better allocation of resources during flood events. Ultimately, this research will contribute to a more sustainable and secure urban planning framework in Nairobi, protecting its population from the devastating impacts of floods.

1.3 Research Identification

The main objective is to compare the effectiveness of two statistical models, Shannon's Entropy (SE) and Frequency Ratio (FR), in predicting flood susceptibility in Nairobi. By evaluating and contrasting these models, the research seeks to determine which method provides more accurate and reliable predictions of flood-prone areas, thereby enhancing flood risk assessment and management in the city. The ultimate goal is to inform and improve urban planning and disaster management strategies to mitigate the impacts of flooding in Nairobi.

1.4 Research Objectives

The specific objectives of the study are:

1. To analyze the influence of flood conditioning factors such as digital elevation model (DEM), land use, soil type, rainfall, and drainage systems on flood susceptibility in Nairobi.
2. To develop flood susceptibility maps using Shannon's Entropy (SE) and Frequency Ratio (FR) models.
3. To compare the prediction performances of SE and FR models in identifying flood-prone areas in Nairobi.
4. Develop a prediction simulation application using the most effective model in GEE

1.5 Justification of the Study

Flooding poses a severe and recurring threat to Nairobi, Kenya, causing widespread socio-economic and environmental impacts. The city's rapid urbanization, inadequate drainage systems, and the effects of climate change have significantly increased its vulnerability to flash floods. Recent events highlight the urgency of this issue: as of mid-March 2024, intense downpours have led to mass flooding across nearly half of Kenya, with Nairobi experiencing extreme levels of inundation. This situation resulted in at least 32 fatalities, the displacement of over 103,500 people, and significant disruptions to daily life, including the closure of roads and suspension of commuter train services.

The Kenyan Red Cross and other emergency services have been overwhelmed by the scale of the disaster, emphasizing the need for more effective flood management strategies. The frequent occurrence of such events and the escalating severity of their impacts underscore the necessity for accurate flood susceptibility mapping to enhance the city's preparedness and resilience.

Accurate flood susceptibility maps are crucial for effective urban planning and disaster management, as they help identify high-risk areas and inform targeted interventions. These maps provide valuable insights into areas at risk of flooding, enabling authorities to implement measures such as improving drainage systems, regulating land use, and developing early warning systems. Such proactive measures can significantly reduce the vulnerability of Nairobi's residents and infrastructure to flooding, promoting a safer urban environment and ensuring better allocation of resources during flood events.

Figure 1:Image by BBC

Figure 2:Image by Climate Change Writers

8 2 LITERATURE REVIEW

2.1 Introduction

Flash floods pose a significant threat to urban areas, especially in rapidly developing cities like Nairobi, Kenya. Effective flood susceptibility mapping is critical for disaster mitigation and urban planning. Various models, including Shannon's Entropy (SE) and Frequency Ratio (FR), have been applied to identify flood-prone areas globally. However, limited research exists on comparing these models in the context of Nairobi's unique hydrological and urban characteristics.

The Frequency Ratio (FR) model is a widely used statistical approach for flood susceptibility mapping due to its simplicity and effectiveness. FR evaluates the spatial correlation between historical flood occurrences and various environmental factors by calculating the ratio of the probability of occurrence to non-occurrence for each factor. This straightforward approach makes FR interpretable and particularly useful in regions with limited data availability. Youssef et al. (2016) applied the FR model to assess flash flood susceptibility in Jeddah, Saudi Arabia, considering factors such as slope, elevation, proximity to rivers, land use, and soil type. The study effectively delineated high-risk areas and produced accurate susceptibility maps. The success of the FR model in this study demonstrates its utility for flood risk assessments, especially in data-scarce environments where more complex models might not be feasible. However, one notable limitation of the FR model is its reliance on the assumption of linearity in the relationship between factors and flood occurrence, which might oversimplify flood dynamics in complex environments.

Shannon’s Entropy on the other hand is a statistical method derived from information theory, initially developed to measure the uncertainty or randomness in a system. In the context of flood susceptibility modeling, SE quantifies the spatial distribution and importance of various flood conditioning factors. This method has been widely applied due to its robustness in handling complex environmental data and its ability to integrate multiple variables effectively. Khosravi et al. (2016) applied the SE model to analyze flash flood susceptibility in Iran, integrating factors such as slope, elevation, land use, soil type, rainfall, and drainage density. Their study demonstrated that SE could accurately capture the spatial variability of flood risks, providing reliable susceptibility maps. The model's strength lies in its ability to identify the most influential factors contributing to flooding, thereby enhancing the understanding of flood dynamics (Khosravi et al., 2016).

One of the most significant gaps lies in the geographic specificity of existing studies. While SE and FR models have been used extensively in Asia and Europe, there is limited research that applies these models to African cities like Nairobi. Nairobi's urban watershed features informal settlements, inadequate drainage systems, and a high degree of impervious surfaces, all of which influence flood dynamics differently compared to the rural or semi-urban areas where these models are typically applied. Another critical limitation is the inadequate integration of urban-specific factors into traditional SE and FR models. In urban areas like Nairobi, factors such as impervious surfaces, high infrastructure density and drainage networks significantly influence flood susceptibility. These variables are rarely included in traditional SE and FR models, which tend to focus on more general environmental factors such as slope, elevation, and proximity to rivers. This oversight can lead to oversimplified assessments that fail to capture the complexities of urban flood dynamics.

A conscious decision of this study is its deliberate incorporation of urban-specific factors such as impervious surfaces, drainage density, and road networks that are frequently underrepresented in traditional flood susceptibility modeling. These variables are critical in urban environments like Nairobi, where dense infrastructure and human interventions heavily influence hydrological processes. For instance, impervious surfaces significantly alter natural infiltration rates, leading to increased surface runoff and localized flooding during heavy rainfall. Similarly, drainage density and road networks not only shape water flow patterns but also determine how effectively floodwaters are channeled or obstructed, making them indispensable in accurately assessing flood susceptibility

Several studies have compared the effectiveness of SE and FR models in flood susceptibility mapping, highlighting their respective strengths and limitations. For instance, Khosravi et al. (2016) compared SE, FR, and other bivariate models in Iran, finding that while both SE and FR performed well, the statistical index model had the highest prediction accuracy. The comparison emphasized that combining SE and FR with other statistical techniques could enhance prediction capabilities.

Another comparative study by Bui et al. (2019) in Vietnam utilized both SE and FR models to assess flash flood susceptibility. The study integrated additional machine learning algorithms to optimize the models, demonstrating that while both SE and FR provided valuable insights, the integration of advanced techniques significantly improved the overall accuracy of the flood susceptibility maps (Bui et al., 2019)

Urban areas present unique challenges for flood susceptibility prediction due to their complex and heterogeneous nature. These complexities include diverse land use patterns (residential, commercial,

industrial, and recreational zones), which interact differently with hydrological processes. High infrastructure density and variability, such as buildings, roads, bridges, and drainage systems, influence flood dynamics, with impervious surfaces increasing surface runoff and higher flood peaks. Micro-topography due to human activities and small-scale elevation variations significantly affect local water flow. Human interventions, such as land reclamation, flood defenses, and modifications to watercourses, further alter natural hydrology. Additionally, data availability and quality pose challenges, as obtaining high-resolution, accurate data on topography, land use, infrastructure, and hydrological conditions is often difficult, limiting prediction accuracy. Studies by Leobons et al. (2019) and Elkhrachy (2015) highlight these complexities, emphasizing the need for robust, detailed data and adaptable models to address urban flood risks effectively

2.2 Remote Sensing and Geospatial Technologies for Flash Flood Analysis

Remote sensing and geospatial technologies have been extensively used for flash flood analysis and prediction. These technologies offer a range of tools for mapping flash flood-prone areas, monitoring flash flood events, and predicting future occurrences. Some of the key technologies used in flash flood analysis and prediction include:

2.2.1 Digital Elevation Models (DEMs)

Digital Elevation Models (DEMs) are digital representations of the earth's surface, which can be used to derive topographic information such as elevation, slope, and aspect. DEMs are widely used for flash flood analysis and prediction as they provide crucial information on the topography of the landscape and the flow of water during flash flood events. Kang et al. (2017) highlighted the importance of DEMs in understanding flash flood dynamics and delineating flash flood-prone areas.

2.2.2 Synthetic Aperture Radar (SAR)

Synthetic Aperture Radar (SAR) is a type of radar imaging that uses microwave energy to detect objects on the earth's surface. SAR is particularly useful for flash flood analysis and prediction as it can provide information on flood extent, water levels, and changes in water dynamics over time. SAR's ability to provide all-weather, day-and-night, high-resolution imagery makes it especially valuable for disaster monitoring. Sentinel-1 SAR has emerged as a pivotal tool in flash flood mapping due to its frequent revisit times and open-access data policy, significantly enhancing flash flood mapping accuracy and timeliness (Smith et al., 2020; Johnson et al., 2019).

2.2.3 Optical Remote Sensing

Optical remote sensing involves the use of visible and near-infrared light to detect objects on the earth's surface. It has been used for flash flood analysis and prediction, providing information on land use, vegetation cover, and water quality. Zhang et al. (2018) demonstrated the effectiveness of optical remote sensing in capturing detailed images of flash flood-affected areas and assessing the impact of flash floods on vegetation and land use.

2.2.4 Google Earth Engine (GEE)

Google Earth Engine (GEE) is a cloud-based platform that provides access to a vast range of satellite and geospatial data, as well as tools for data analysis and visualization. GEE has been widely used for flash flood analysis and prediction due to its ability to process and analyze large volumes of satellite data quickly and efficiently. Zhang et al. (2020) emphasized GEE's role in facilitating large-scale environmental monitoring and real-time flash flood analysis.

2.2.5 Geographic Information Systems (GIS)

Geographic Information Systems (GIS) are computer-based tools for capturing, storing, analyzing, and visualizing geospatial data. GIS has been extensively used for flash flood analysis and prediction, providing a range of tools for mapping flash flood-prone areas, analyzing flash flood risk, and predicting flash flood events. Yu et al. (2018) highlighted GIS's capabilities in integrating various data sources to create comprehensive flash flood risk assessments.

2.2.6 R

R is a programming language and software environment widely used for statistical computing and graphics. In flash flood analysis, R offers numerous packages for spatial data manipulation, analysis, and visualization. Packages such as "sp", "raster", and "sf" facilitate the handling of spatial data, while "ggplot2" and "leaflet" enable detailed and interactive visualizations. The versatility of R in processing large datasets and performing complex statistical analyses makes it a valuable tool in flash flood prediction and risk assessment. Additionally, R can integrate with GIS and remote sensing data to enhance flash flood modeling efforts.

9 3 MATERIALS AND METHODS

3.1 Study Area

The Nairobi watershed, located within the capital city of Kenya, lies in the larger Athi River Basin, stretching approximately between 1°10′ S and 1°30′ S latitude and 36°40′ E and 37°00′ E longitude (Figure 1). This watershed encompasses an area of about 3,212 square kilometers, playing a crucial hydrological role in the region. Nairobi city serves as the administrative and economic hub of Kenya, and the watershed is home to a dense population and dynamic land-use patterns. The elevation in the Nairobi watershed ranges from 1,471 meters to 1,927 meters above sea level. The Nairobi watershed is a sub-catchment of the larger Athi River Basin, one of Kenya's major drainage systems, eventually draining into the Indian Ocean. The watershed is primarily defined by the Nairobi River and its key tributaries, including the Mathare, Ngong, and Mbagathi Rivers. These rivers originate from the surrounding highlands and flow through Nairobi, shaping its hydrological system. The Mathare River drains the northern parts of the watershed, the Ngong River originates from the Ngong Hills to the west, and the Mbagathi River forms the southern boundary of the watershed, all contributing to the Athi River. The interconnectedness of these rivers with the Athi River Basin underscores their role in Nairobi’s flood dynamics.

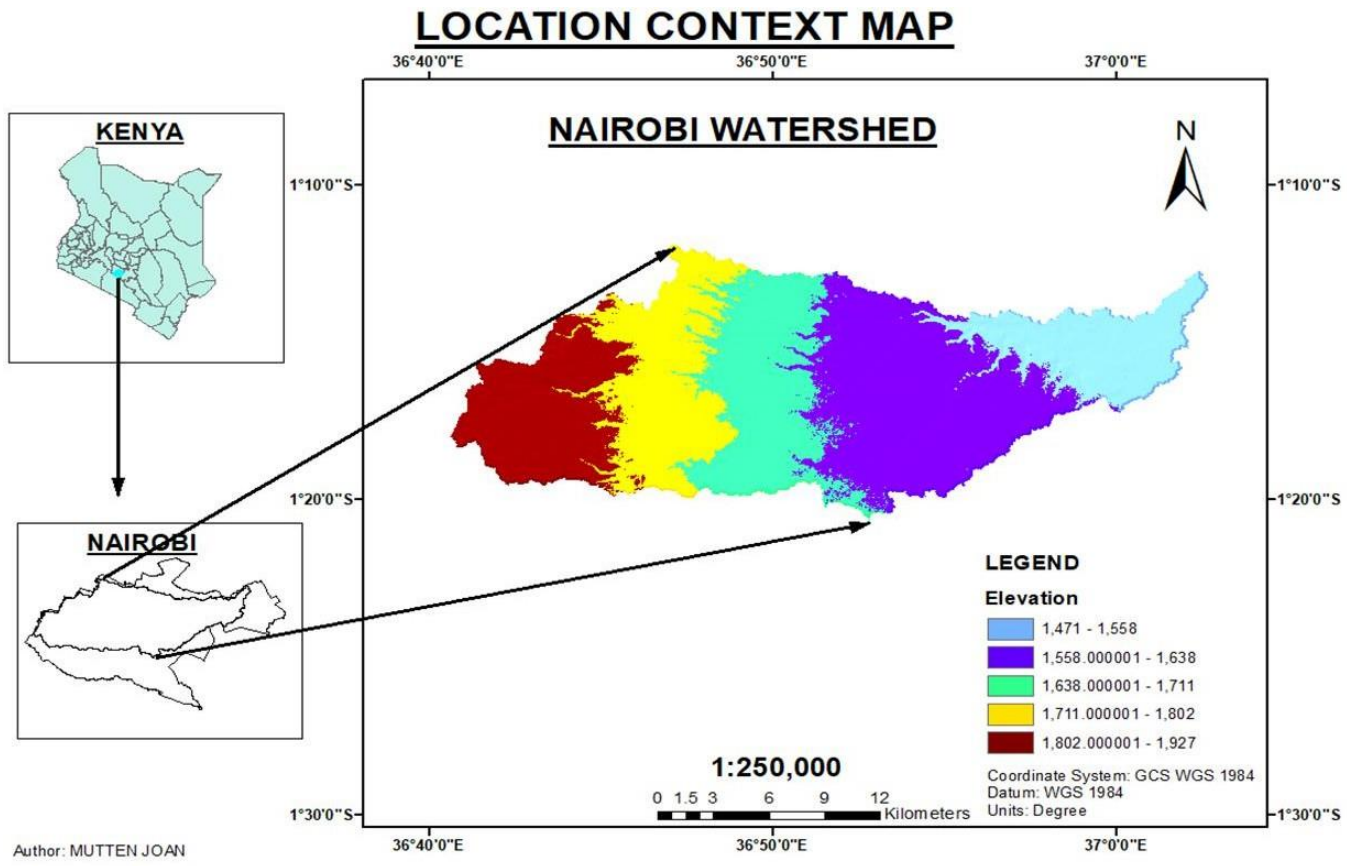


Figure 3:Study Area

3.2 Data

Table 1:Data

Data	Source	Resolution	Link
Sentinel 2	European Space Agency	10m	scihub.copernicus.eu
Sentinel 1 SAR			
SRTM DEM	United States Geological Survey (USGS)	30	earthexplorer.usgs.gov
Soil Map	IGAD Climate Prediction and Applications Centre (IPAC)		IPAC
Road density	Humanitarian Data exchange	30	data.humdata.org
River density	SRTM DEM	30	earthexplorer.usgs.gov
Drainage density	Open Street Map (OSM)	30	
NDVI	European Space Agency	30	scihub.copernicus.eu
LULC			
Aspect	United States Geological Survey (USGS)	30	earthexplorer.usgs.gov
Slope			
Plane Curvature			

In the current study, Sentinel-1 and Sentinel-2 satellite images were used in the GEE cloud platform. In the GEE platform, all of the preprocessing, such as Radiometric and Atmospheric corrections, which have already been performed, reduce the essential time needed to prepare the data for the main analysis.

3.2.1 Sentinel-1

The S1 mission comprises a constellation of two polar-orbiting satellites (Sentinel-1A and Sentinel-1B) that operate day and night, performing C-band synthetic aperture radar (SAR) imaging, which enables them to acquire imagery regardless of the weather and achieve global coverage every 6 days. Sentinel-1 SAR aids in flood extent delineation, floodwater characterization, and mapping of inundated areas. The frequent revisit times and open-access data policy of the Sentinel-1 mission significantly enhance flood mapping accuracy and timeliness, supporting disaster management and mitigation efforts.

3.2.2 Sentinel-2

The S2 mission comprises a constellation of two polar-orbiting optical satellites placed in the same orbit (Sentinel-2A and Sentinel-2B), phased at 180◦ to each other. Sentinel-2A was launched on 23 June 2015, and Sentinel-2B on 7 March 2017. Sentinel-2 data enables the identification of flood extent, floodwater characteristics, and flood-induced changes in land cover, which are crucial for flood mapping and post-flood damage assessment.

3.2.3 SRTM DEM

The Shuttle Radar Topography Mission Digital Elevation Model (SRTM DEM) is a comprehensive and highly detailed global topographic dataset created using radar technology. Launched in February 2000 as a collaborative effort between NASA and other international space agencies, SRTM employed a radar instrument on the Space Shuttle Endeavour to capture elevation data of Earth's surface. By providing precise topographic data at a global scale, SRTM DEM enables the accurate delineation of terrain features critical for flood assessment. It plays a central role in floodplain modeling by allowing researchers to identify flood-prone areas, simulate flood scenarios, and estimate floodwater depths with a high degree of precision.

3.3 Methodology

The methodology utilized for this study is summarized in the flowchart below;

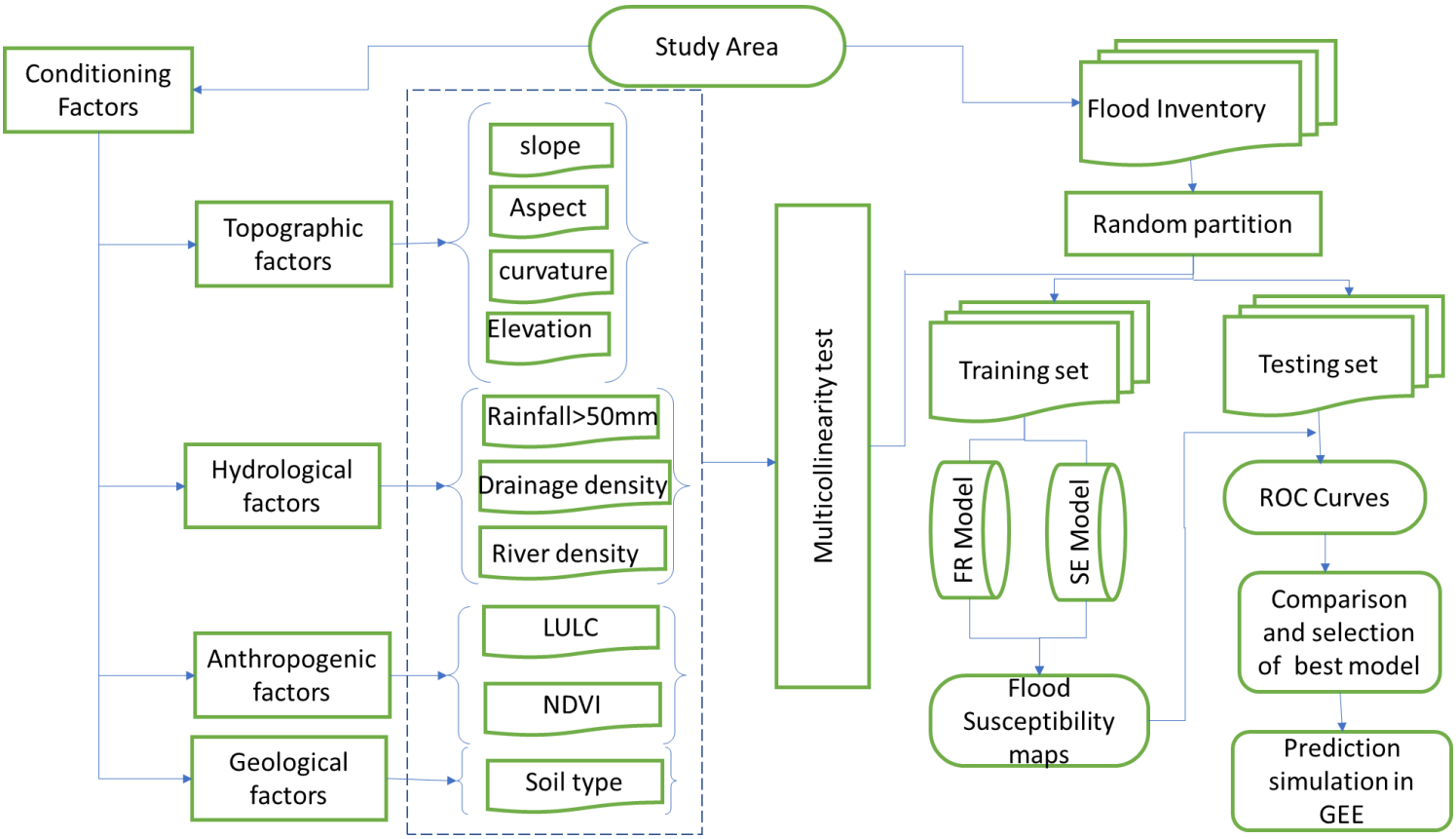


Figure 4:Methodology flowchart

3.3.1 Flood inventory map using GEE

An accurate flood inventory map is essential in flood studies, as it significantly impacts the precision of flood susceptibility modeling (Choubin et al. 2019). This process involves identifying the locations of historical flood events and creating an inventory map to aid in accurate flood susceptibility modeling (Tehrany et al. 2013).

I conducted the following steps:

Satellite Data Selection and Preprocessing

1. Satellite Data Selection:

Selected Sentinel-1 SAR data from the COPERNICUS/S1_GRD Image Collection.

Filtered for 'VH' polarization and 'IW' instrument mode.

Included both ascending and descending orbit properties.

2. Filtering Data by Date and Region:
    - Filtered the dataset for dates before the flood event from January 1, 2024, to January 31, 2024, and for dates after the flood event from April 19, 2024, to May 25, 2024.
3. Creating Mosaics and Clipping to Study Area:
    - Created mosaic images of the filtered collections and clipped them to the study area.
4. Applying Speckle Filtering:
    - Applied Refined Lee speckle filtering to the before and after images to reduce noise and enhance the quality of the SAR images.

Identification of Inundated Pixels

1. Difference Calculation:

Calculated the difference between the after and before filtered images to identify changes due to flooding.

2. Flood Detection:

Set a threshold for identifying flood pixels and created a flood mask to delineate the flooded areas.

3. Water Body Detection:

Identified persistent water bodies in both the before and after images to distinguish them from transient floodwaters.

Visualization and Analysis

1. Mapping:

Centered the map on the study area and added layers for visualization, including filtered images, flood inundation, and water bodies.

2. Flood Area Calculation:

Calculated the total flooded area within the district using the flood mask and determined the extent of the flooding in hectares.

Export

1. Export Flood Map:

Exported the flood map to Google Drive for further analysis and use in subsequent stages of the project.

Image ID/Source	Event Time	Acquisition Date	Processing Level	Spatial Reference
COPERNICUS/ S1_GRD (Before)	Pre-flood	2024-01-01 to 2024-01-31	IW Mode	Datum: WGS84, EPSG:4326
COPERNICUS/ S1_GRD (After)	Post-flood	2024-04-19 to 2024-05-25	IW Mode	Datum: WGS84, EPSG:4326

Table 2:Sentinel-1 SAR Data Acquisition Details for Flood Analysis: Pre- and Post-Flood Observations

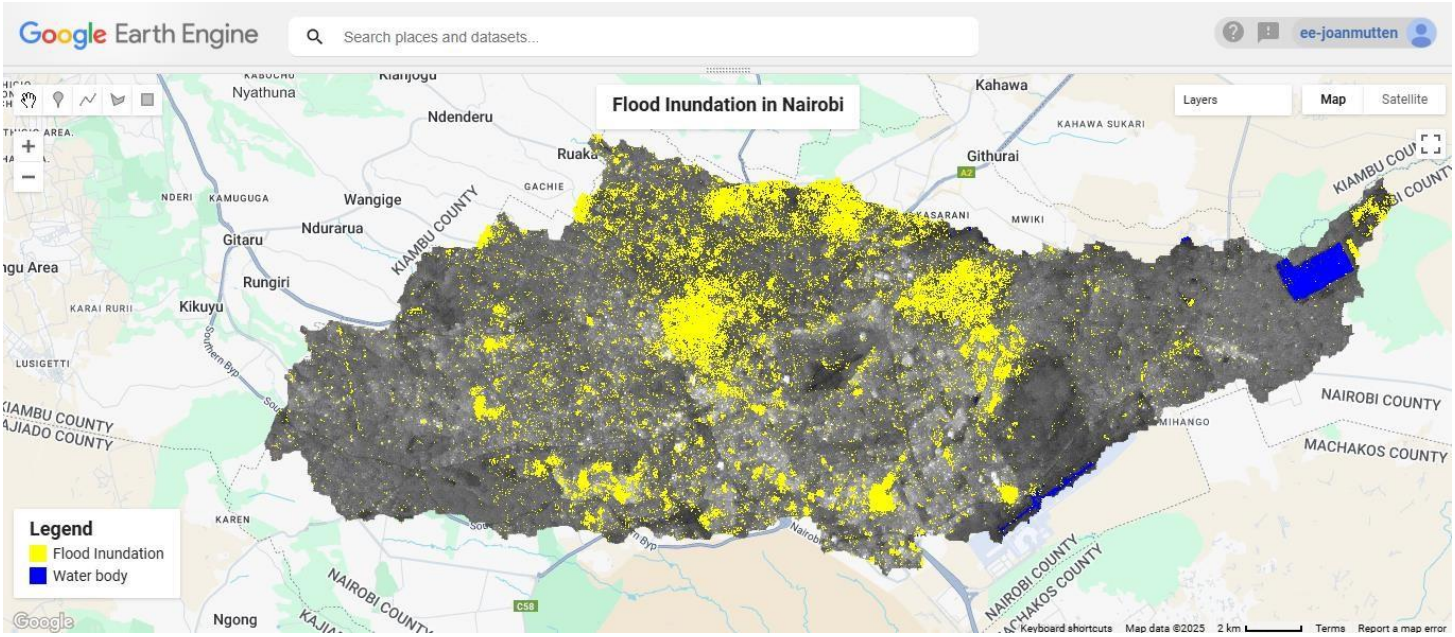


Figure 5:Flood Map GEE

3.3.2 Flood Inventory processing in R

The flood inventory analysis was performed using a flood map of the Nairobi River Basin, created in Google Earth Engine (GEE) and processed in R. The flood map, with a spatial resolution of 30 meters by 30 meters, provided a classification of the region into flooded and non-flooded areas. This analysis aimed to quantify the extent of flooding and non-flooded areas to support flood risk management efforts in the basin.

Steps followed

The analysis began with loading the GeoTIFF file of the flood map into R using the terra package. The flood map contained two classes: flooded areas, represented by a value of 1, and non-flooded areas, represented by a value of 0. To determine the spatial extent of these classes, the pixel counts for each category were extracted using the freq() function.

Each pixel in the flood map corresponds to an area of 900 square meters (30m × 30m). The total area for each flood class was calculated by multiplying the pixel count by 900. These areas were then converted to square kilometers by dividing the result by 1,000,000. The calculated areas were documented for further use in understanding the extent of flooding in the Nairobi River Basin. Additionally, the results were saved in a CSV file for reference and integration into the overall project documentation.

The flood inventory analysis revealed that the Nairobi River Basin encompasses a total area of 670.023 square kilometers. Of this, 612.684 square kilometers (approximately 91.44%) were classified as non-flooded, while 57.339 square kilometers (approximately 8.56%) were identified as flooded. The pixel counts for the non-flooded and flooded areas were 680,760 and 63,710, respectively.

These findings highlight the spatial extent of flooding within the basin. While the majority of the region remains unaffected by flooding, the flooded areas represent significant zones that may pose risks to local communities, infrastructure, and ecosystems.

Flood Class	Pixel Count	Area (m^2)	Area (km^2)
Non-flooded (0)	680,760	612,684,000	612.684
Flooded (1)	63,710	57,339,000	57.339

Table 3:Flood Classification Results: Pixel Count and Area of Flooded and Non-Flooded Regions

Total Analyzed Area: 670.023 km^2

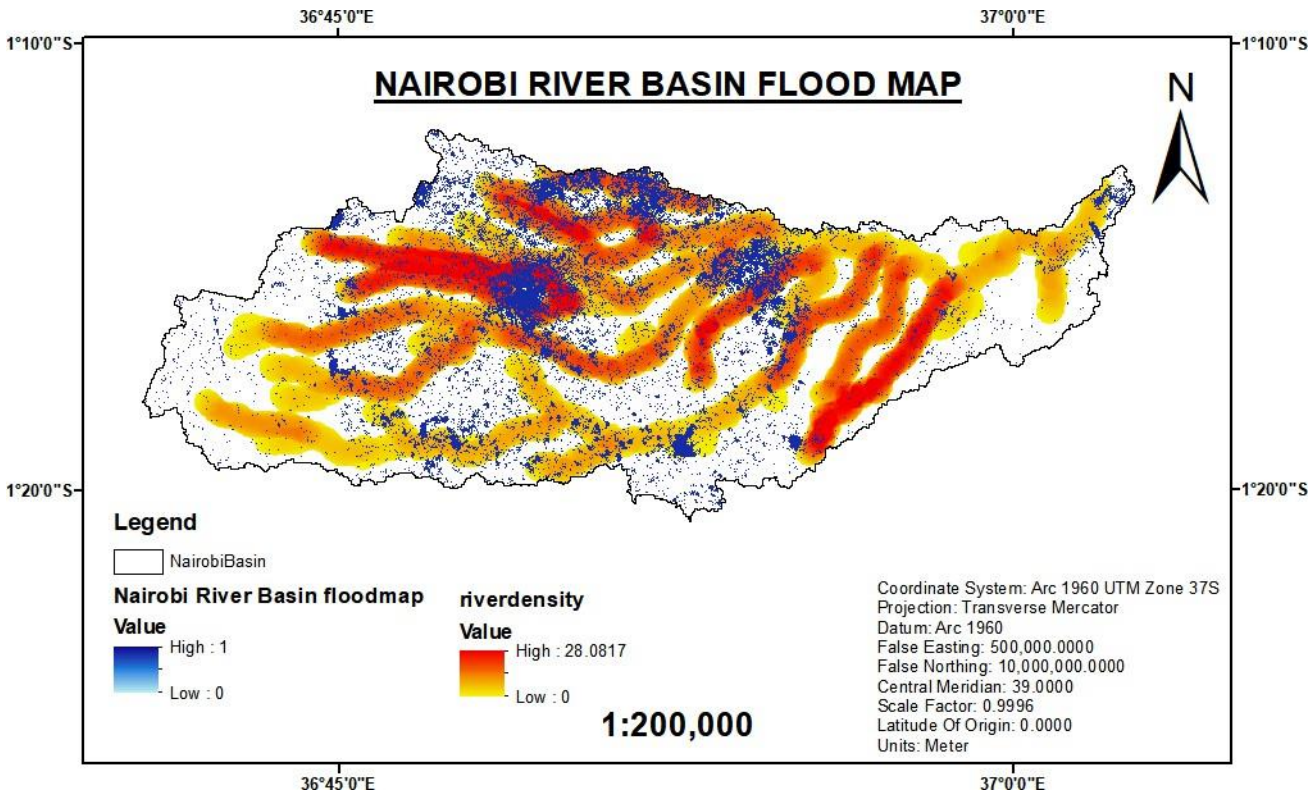


Figure 6:Flood distribution and river density

3.4 Flood conditioning factors

3.4.1 Slope Angle

Slope refers to the change in angle from the horizontal plane or the change in height per unit distance traveled. It is a measure of the steepness of a line or plane and is expressed in percentage or degrees. Slope has a direct effect on surface runoff and the degree of water infiltration into the ground, making it one of the critical explanatory factors in flood occurrence prediction studies (Tehrany et al. 2019). Steeper slopes result in faster runoff and reduced infiltration, which can increase the likelihood of flash floods in lower-lying areas. Conversely, gentler slopes may allow for more infiltration, reducing the immediate impact of heavy rainfall.

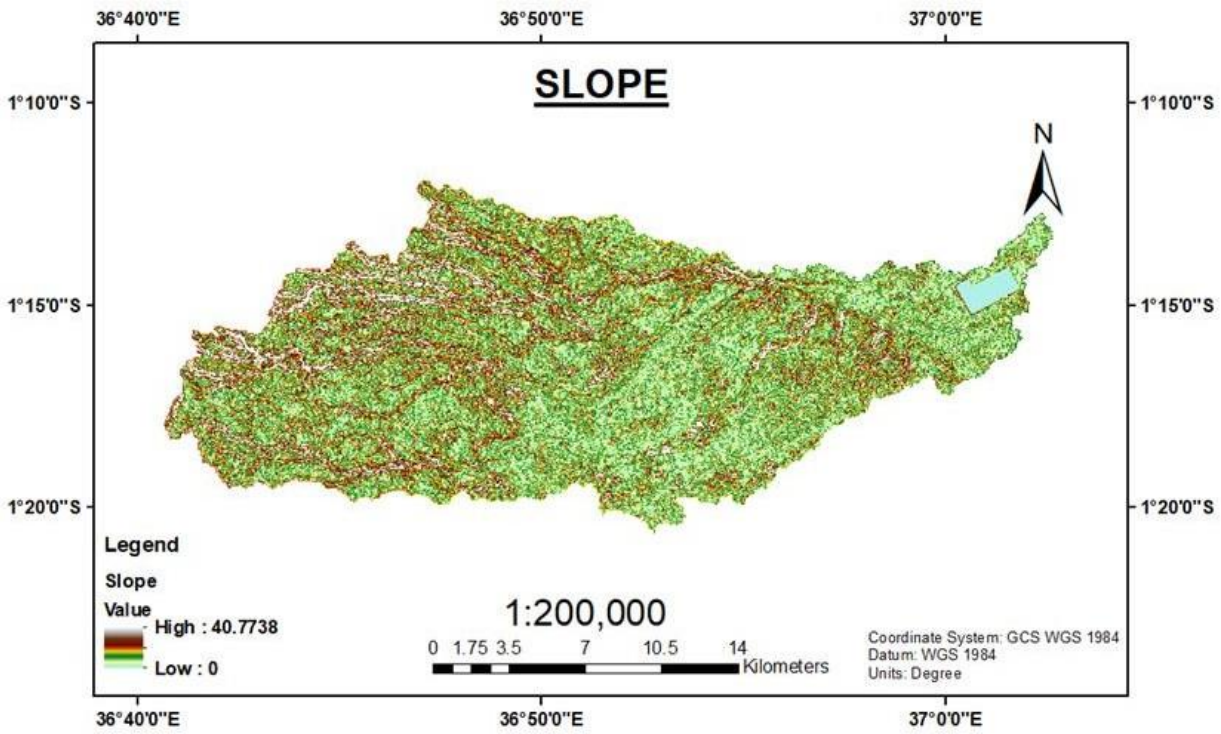


Figure 7:slope map

3.4.2 Slope Aspect

Slope direction has a strong influence on hydrological processes (especially through frontal precipitation direction and evapotranspiration), and as a result it also affects vegetation, even more so in dry environments (Sidle & Ochiai 2006). Slope aspect is defined as the direction in which the slope of a surface faces. It is a determinant factor for the occurrence of sunlight on the surface, which affects the characteristics of rainfall, soil moisture, and vegetation growth (Zhou and Liu 2004; Burnett et al. 2008; Gallardo-Cruz et al. 2009).

For the study area, the topographic surface has been classified into ten aspect categories to understand the relationship between slope direction and flash flood susceptibility. This classification helps in identifying which slopes are more likely to contribute to rapid runoff and potential flooding.

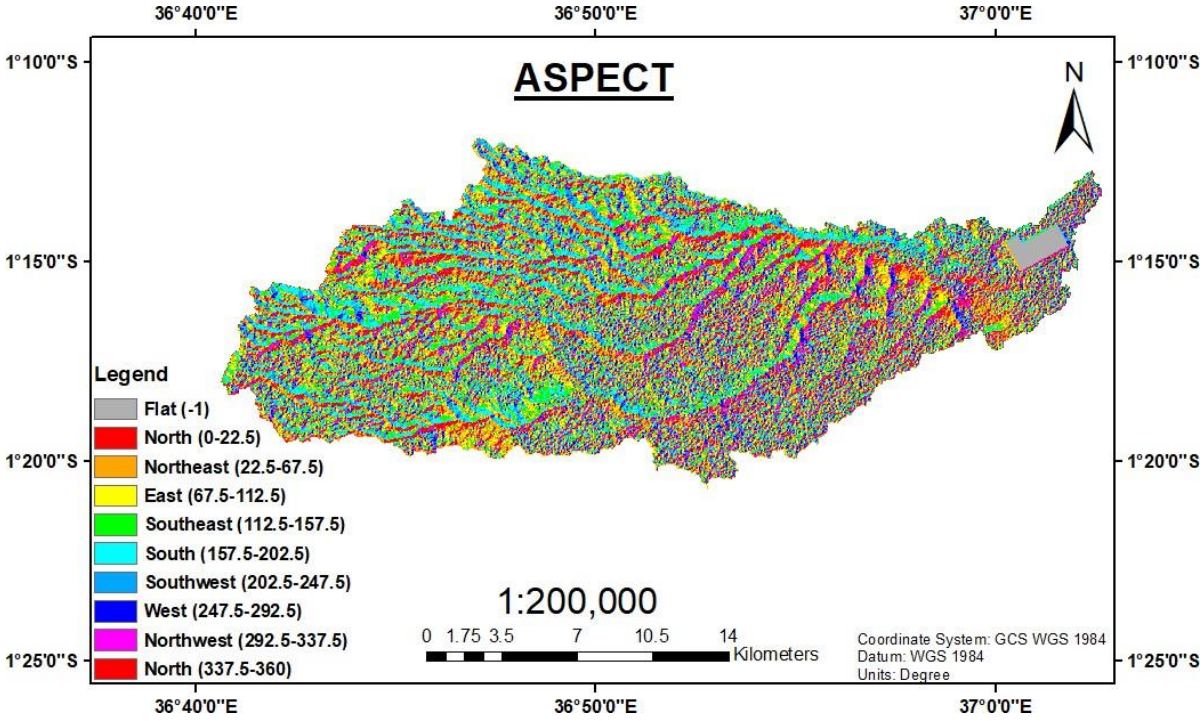


Figure 8:Aspect map

3.4.3 Plan Curvature

This parameter is indicative of processes related to accumulation, flow velocity, and erosion (Mirzaei et al. 2021). Curvature affects the flow and it also influences the occurrence of floods (Pradhan 2010), and floods mostly occur in areas with flat curvature (Tehrany et al. 2015a, 2015b). Plan curvature, or planform curvature, refers to the curvature of the terrain in the direction parallel to the maximum slope (Kimerling et al. 2016). It is an important factor in understanding how water flows across the land surface. Positive values of plan curvature indicate convex surfaces, negative values indicate concave surfaces, and zero values indicate flat or planar surfaces. This is particularly useful in flood studies as it can indicate areas where water is likely to accumulate or disperse. For example, concave areas may collect water, increasing flood risk, while convex areas may shed water, reducing flood risk.

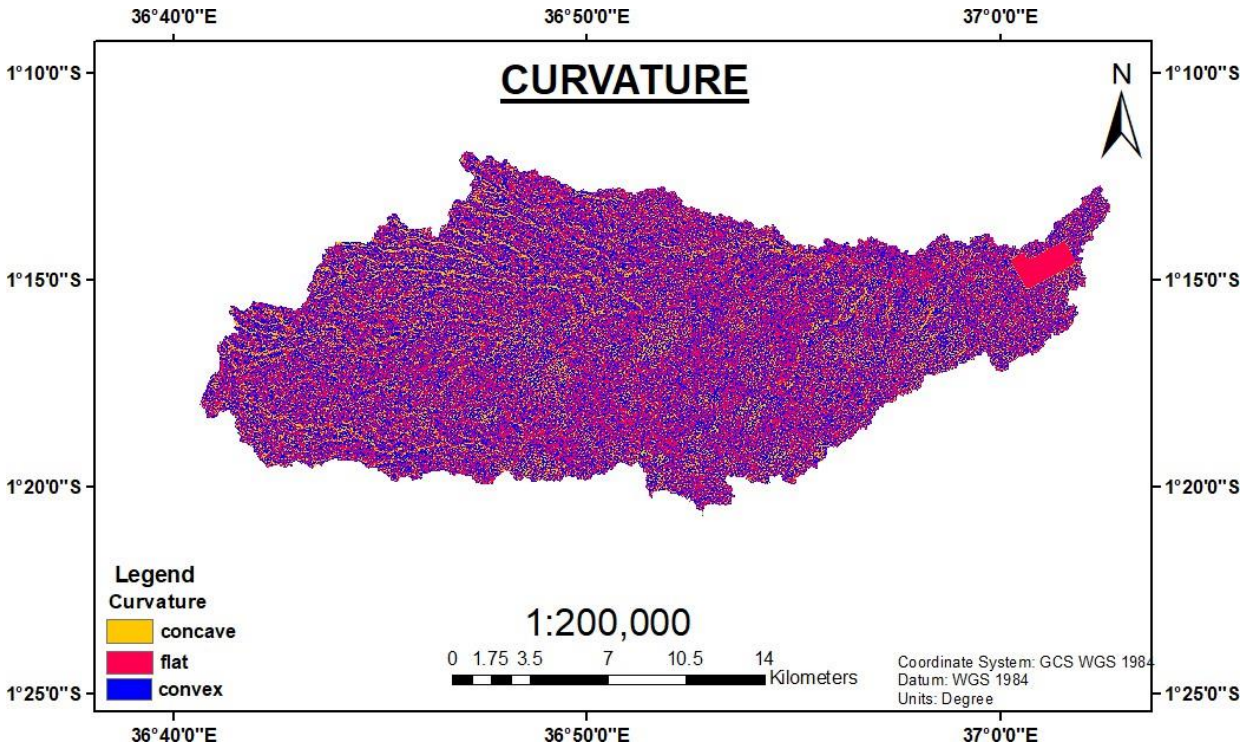


Figure 9:Curvature map

3.4.4 Altitude

In the Nairobi watershed, elevation plays a significant role in influencing flood susceptibility. The elevation affects how water flows across the landscape. Low-lying areas, particularly those near rivers, are more prone to flooding due to the accumulation of water from higher elevations. In contrast, steep areas may experience faster runoff, which can contribute to flash floods downstream. Furthermore, changes in elevation can influence the direction and intensity of water flow, potentially overwhelming drainage systems in urban areas, thus exacerbating flood risks.

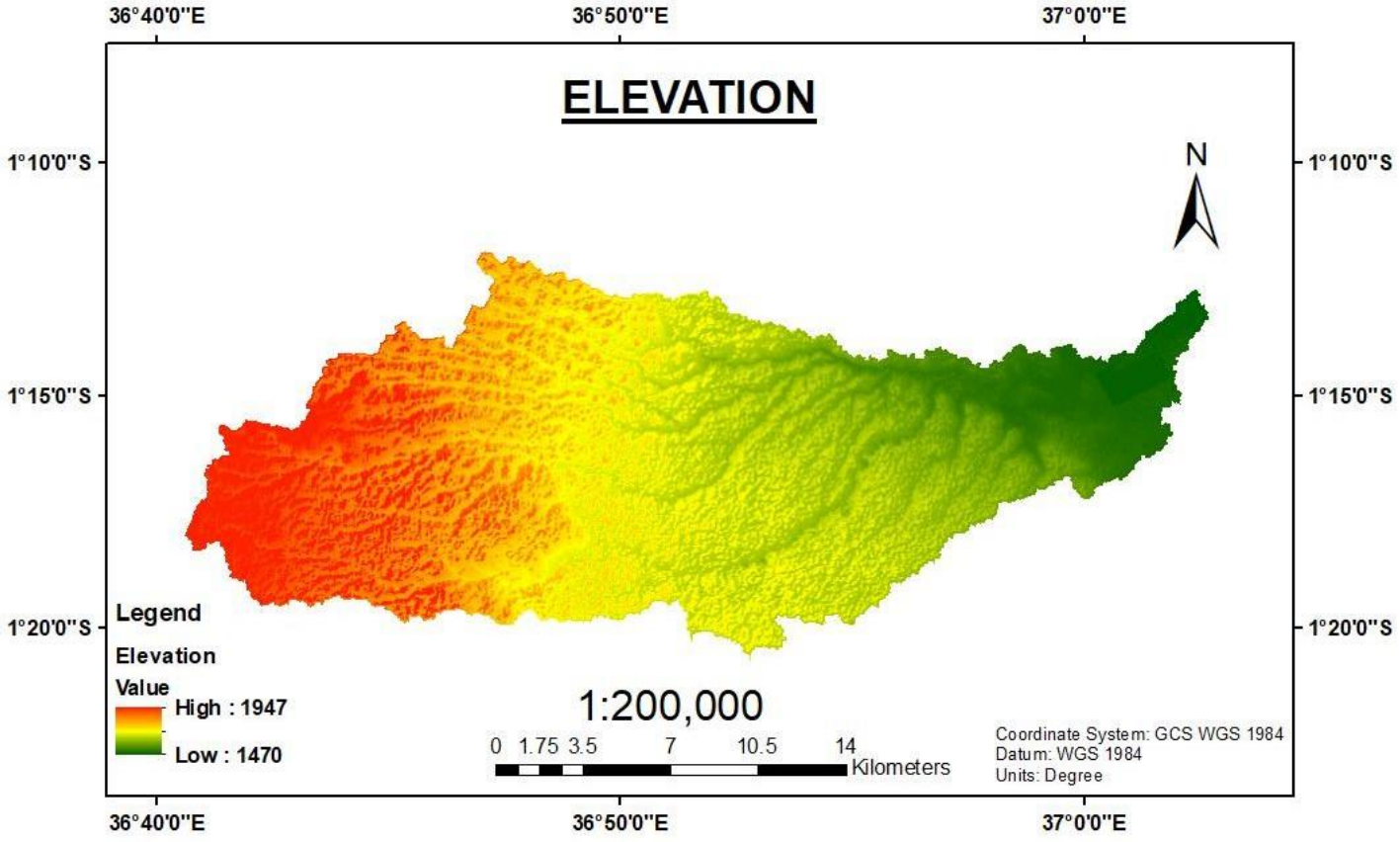


Figure 10:Elevation map

3.4.5 River Density

The river density data used in this flood susceptibility analysis was derived from a DEM by delineating river networks using a flow accumulation threshold to identify hydrologically significant streams while excluding minor channels. The extracted river network was then processed using the Line Density tool in ArcMap, calculating river density within a moving window to create a spatially varying representation of river concentration across the study area.

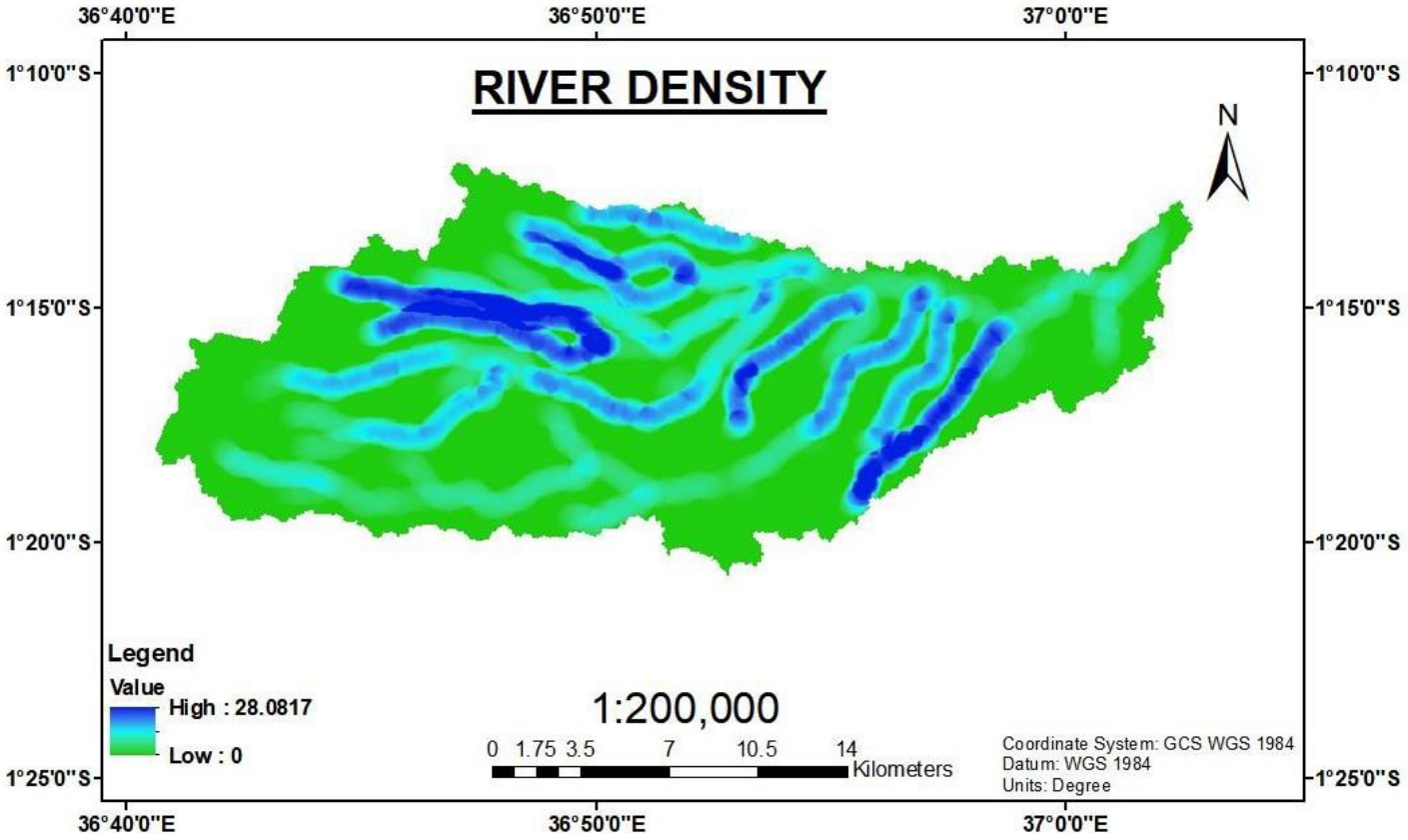


Figure 11:River Density

3.4.6 Drainage Density

Drainage density is the ratio of total length of watershed channels to the area of the basin. Waterways network in watershed shows the way water flows in the basin (Abdideh et al. 2011; Tahmassebipoor et al. 2016). It serves as an important indicator of how well a region is drained by its river network and has a direct impact on flood susceptibility. High drainage density indicates a well-developed network of rivers and streams that may quickly collect and channel runoff during rainfall. However, this can also lead to an increased flood risk, particularly in areas where the channels overflow due to the inability to carry large volumes of water during intense rain events. Conversely, low drainage density may imply fewer channels to drain the water, leading to slower runoff and the possibility of water accumulating in flat, low-lying areas, which increases the potential for flooding.

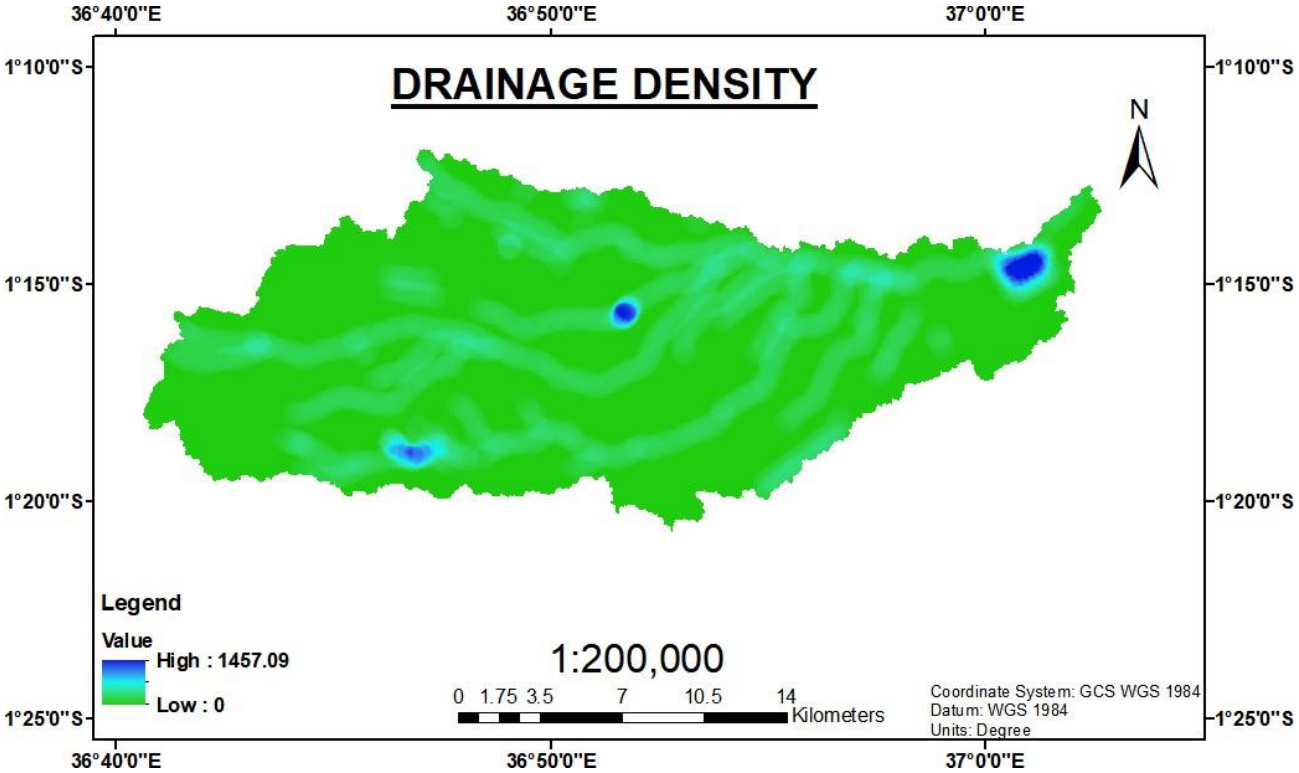


Figure 12:Drainage density

3.4.7 Distance from Rivers

The spread of the flood in the basin is related to the distance of the location to the drainage system (Elkhrachy 2015). While less flooding occurs in areas far from the drainage system, it occurs more in closer areas (Samanta et al. 2018). In this respect, drainage network density is one of the important parameters for flood sensitivity analysis. Distance from river decides how intense the impact of flood will be and up to what distant place (from river) the flood water will spread. Therefore, flood spread and magnitude are directly influenced by this explanatory factor (Glenn et al. 2012). DEM data have been used for extraction of streams. Then vectorized streams have been used as input through Euclidian distance algorithm to compute distance from streams. The final 'distance from streams' map was generated in raster format and resampled at 30 m spatial resolution.

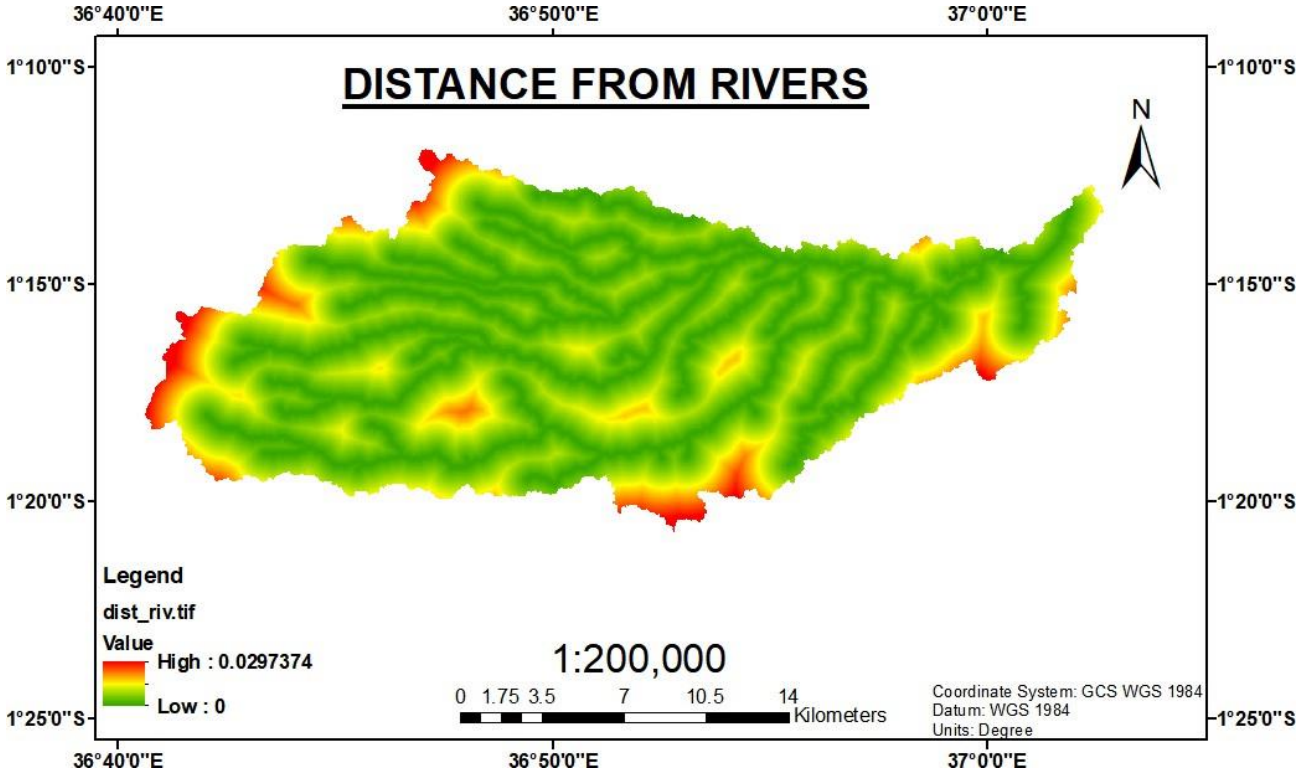


Figure 13:Distance from rivers

3.4.8 Road Density

Road density influences flooding by affecting how water flows and accumulates in urban areas. Road data was obtained from OpenStreetMap, then rasterized, and the density was calculated using the Line Density tool in ArcMap. The total length of roads within a specified radius was used to generate a road density map in raster format. This map, resampled to 30m spatial resolution.

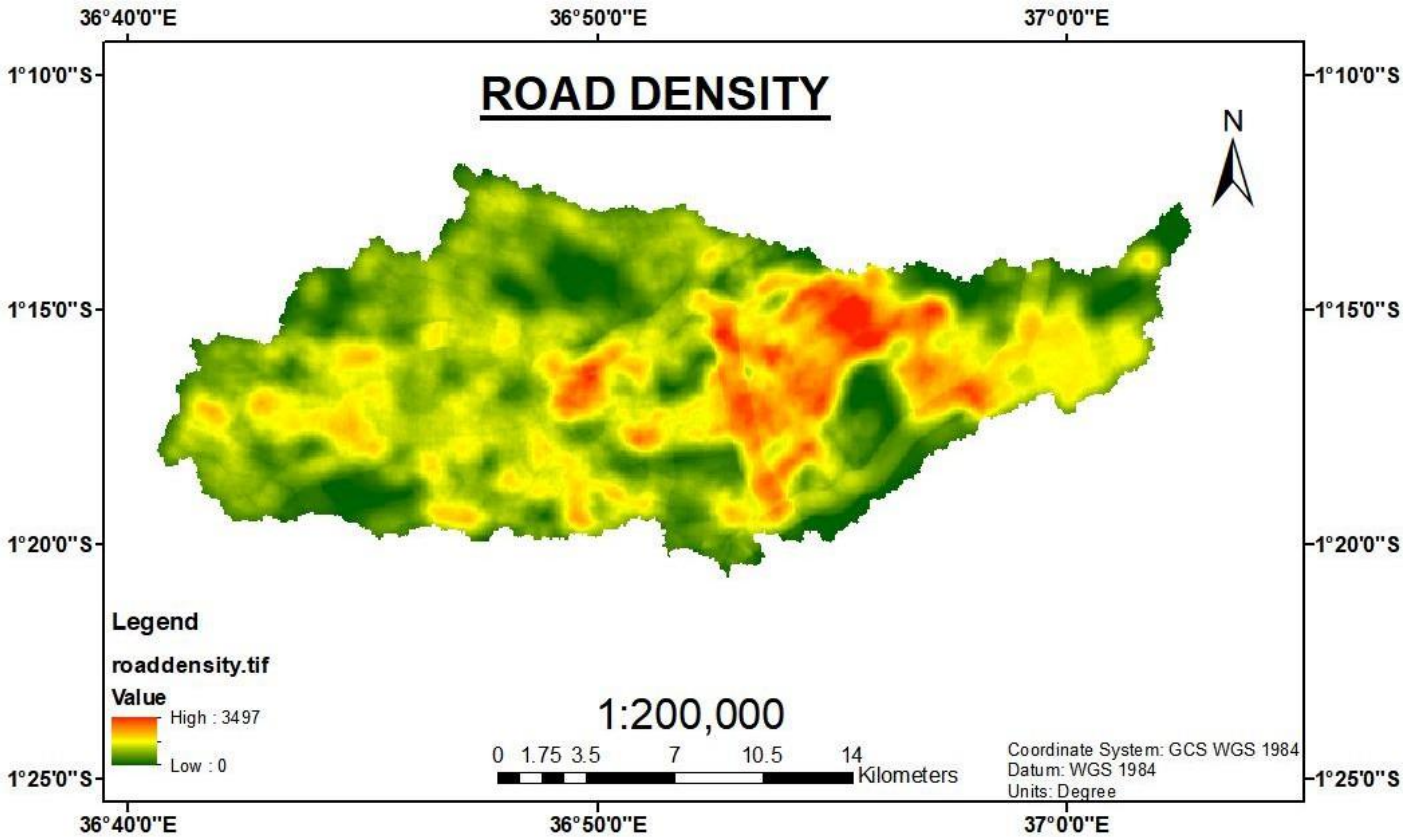


Figure 14:Road Density

3.4.9 LULC

Land use/land cover of an area denotes natural as well human induced cover of the earth's surface. Land use types, directly or indirectly dictate many hydrological process components such as run off generation, evapotranspiration, infiltration, etc. To use this dataset into different models of flood susceptibility, the subset data with the five major categories of land use/land cover (LULC)-vegetation, forest, built-up land, bare land and waterbodies (rivers/lakes).

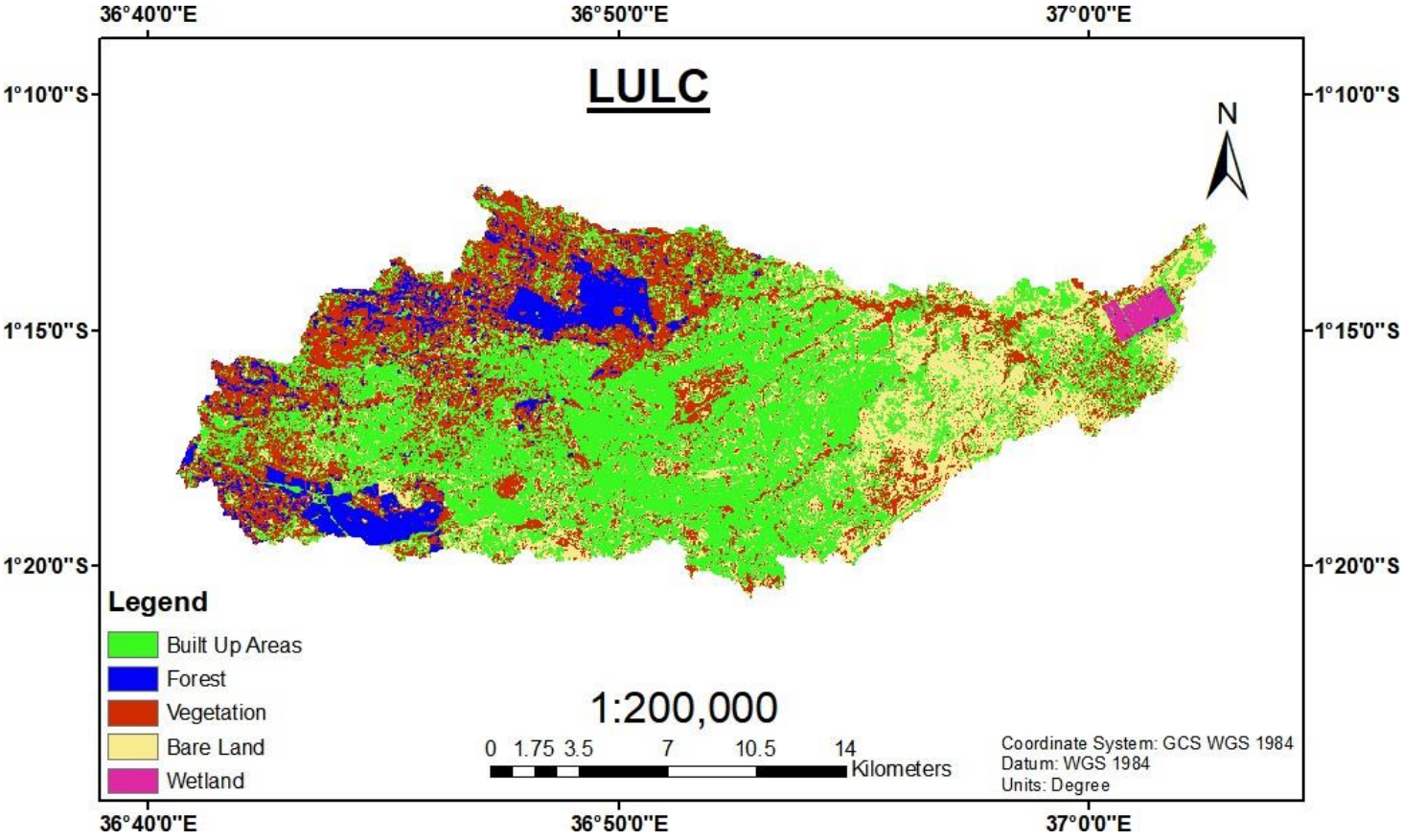


Figure 15:Land Use Land Cover

3.4.10 NDVI

The range of NDVI index values varied between +1 and −1 that show a measure of vegetation characteristic of an area. The Normalized Difference Vegetation Index (NDVI) is a key conditioning factor in flood susceptibility mapping, reflecting the density and health of vegetation cover, which significantly influences hydrological processes. NDVI affects infiltration rates, surface runoff, and soil erosion, making it a crucial parameter for assessing flood risk. Regions with lower NDVI values, indicating sparse vegetation or bare soil, are more susceptible to flooding due to reduced infiltration and increased runoff. Conversely, areas with higher NDVI values, characterized by dense vegetation, tend to mitigate flood risks by enhancing water absorption and reducing surface runoff.In this analysis, NDVI was generated using Google Earth Engine (GEE) from multispectral Landsat satellite imagery, providing high-resolution data on vegetation cover.

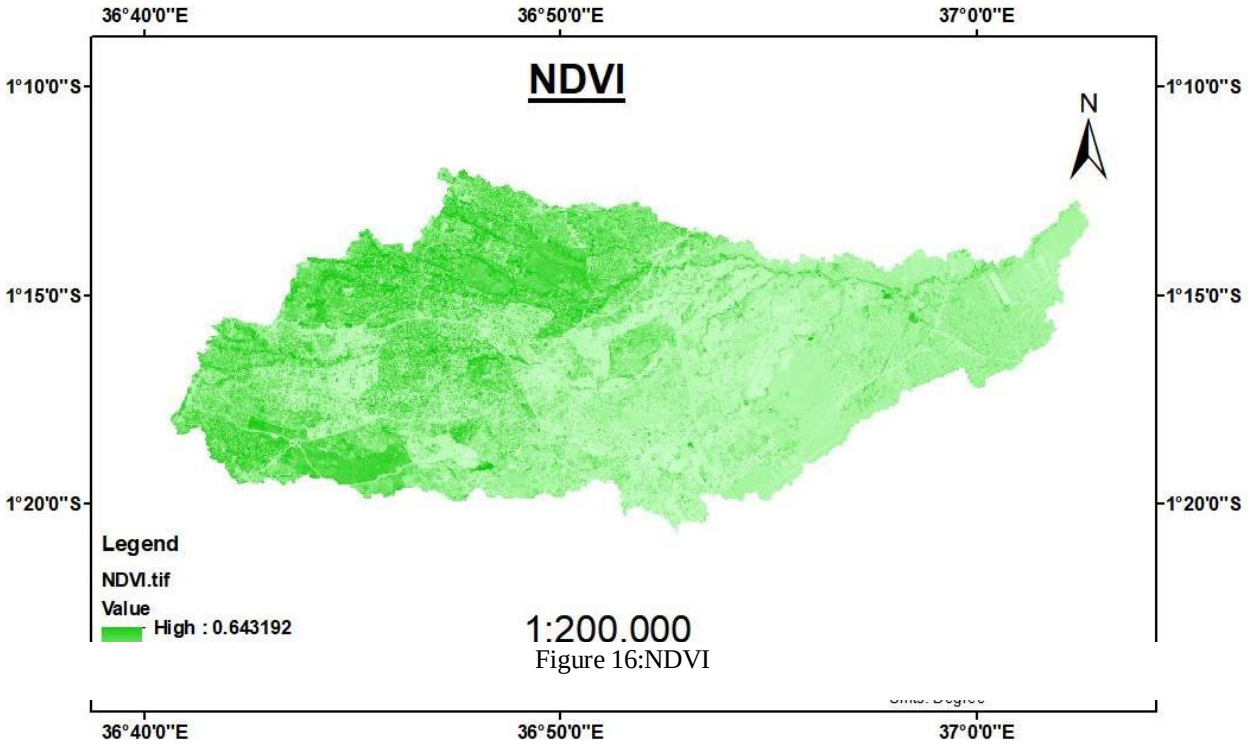


Figure 16:NDVI

3.4.11 Rainfall

The most important parameter in the occurrence of floods is rainfall (Tehrany et al. 2015a, 2015b; Khosravi et al. 2018). The time–space relationships of precipitation are complex; however, they are the basic calculations for flood simulation models (Khosravi et al. 2018). The intensity and duration of the precipitation, as well as the amount of precipitation, are also effective for the development of the flood and flood duration, its area of influence, and possible damages to occur in the region. Heavy downpour of rainfall is the cause of floods in a very large number of cases. Annual rainfall above 50mm data used in this study has been derived from Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS). The raster data which was initially at 0.05 degrees, which is approximately 5.5 kilometers at the equator resolution has been resampled using nearest neighbor method to comply homogeneity with other datasets used.

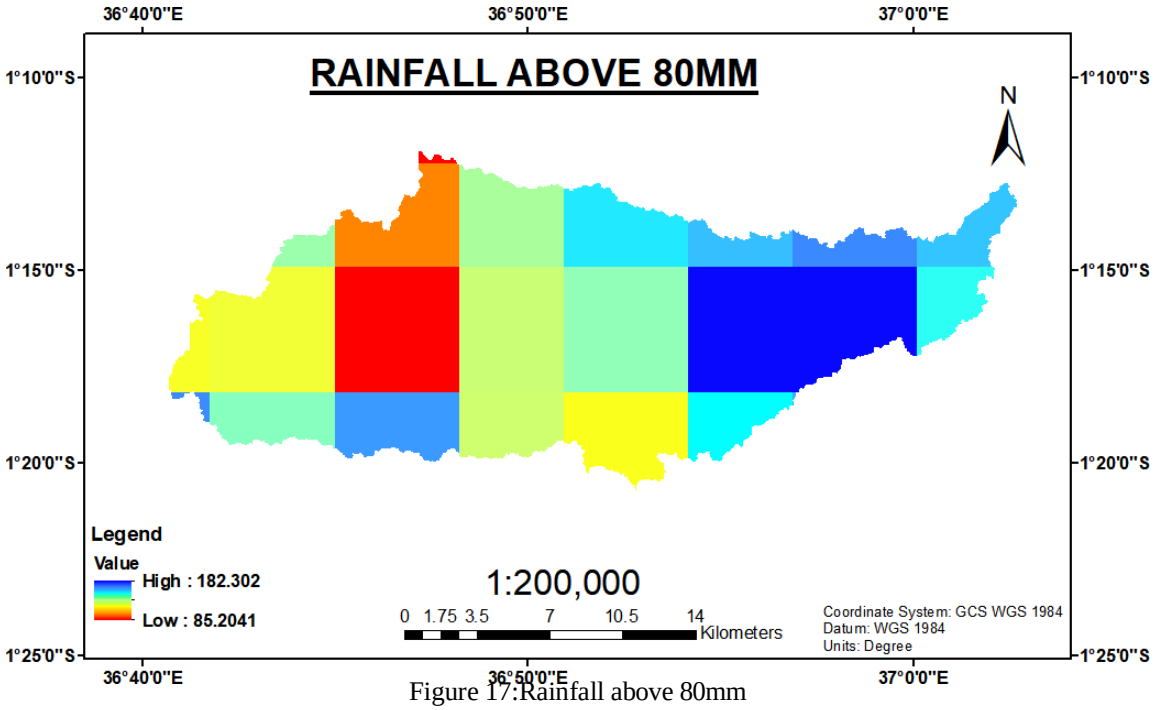


Figure 17:Rainfall above 80mm

3.4.12 Soil

Many of the soil properties i.e. soil texture, soil type, etc. affect water infiltration into the ground and available soil moisture affects how fast or slow a surface covered with a definite soil type will be drained during the floods (Sangati et al. 2009; Unger et al. 2009). Soil database used in the study has been downloaded from IGAD Climate Prediction and Applications Centre (IPAC) soil data portal. The available data has been rasterized using the soil drainage field and then resampled to 30m spatial resolution to make it comparable to other datasets and to be used as input in the models.

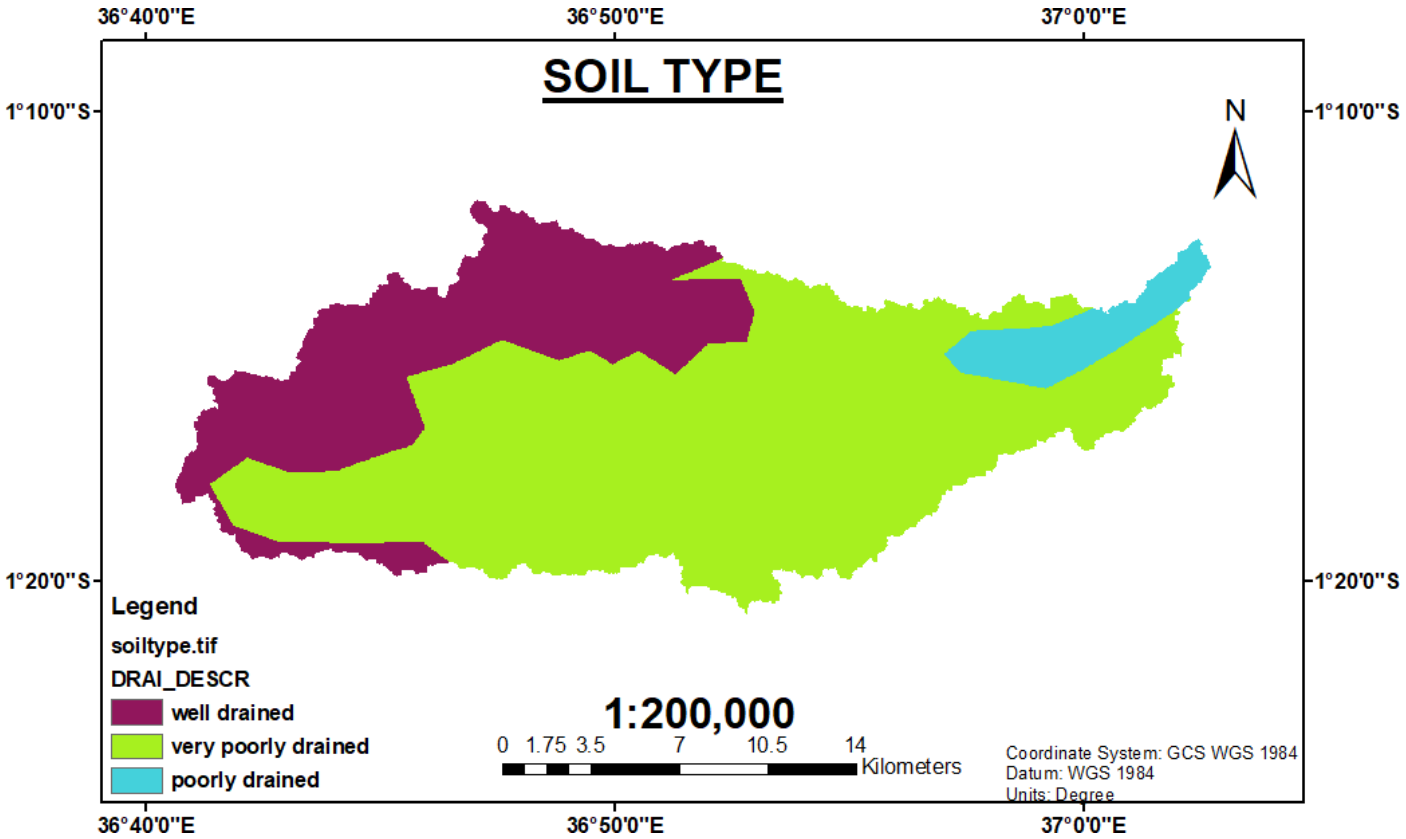


Figure 18:Soil types

3.5.1 Correlation matrix

The correlation matrix is an essential exploratory tool used to identify pairwise relationships between variables in a dataset. It provides numerical values, ranging from -1 (perfect negative correlation) to 1 (perfect positive correlation), to indicate the strength and direction of these relationships. By visualizing the correlations, I can identify variables that are highly correlated (e.g., r>0.8r > 0.8r>0.8 or r<−0.8r < -0.8r<−0.8), suggesting redundancy or overlapping information. The correlation matrix helps in the initial stages of variable selection by pointing out these dependencies, enabling to decide which variables might be redundant or unnecessary. The correlation coefficients between the considered 12 factors is shown below.

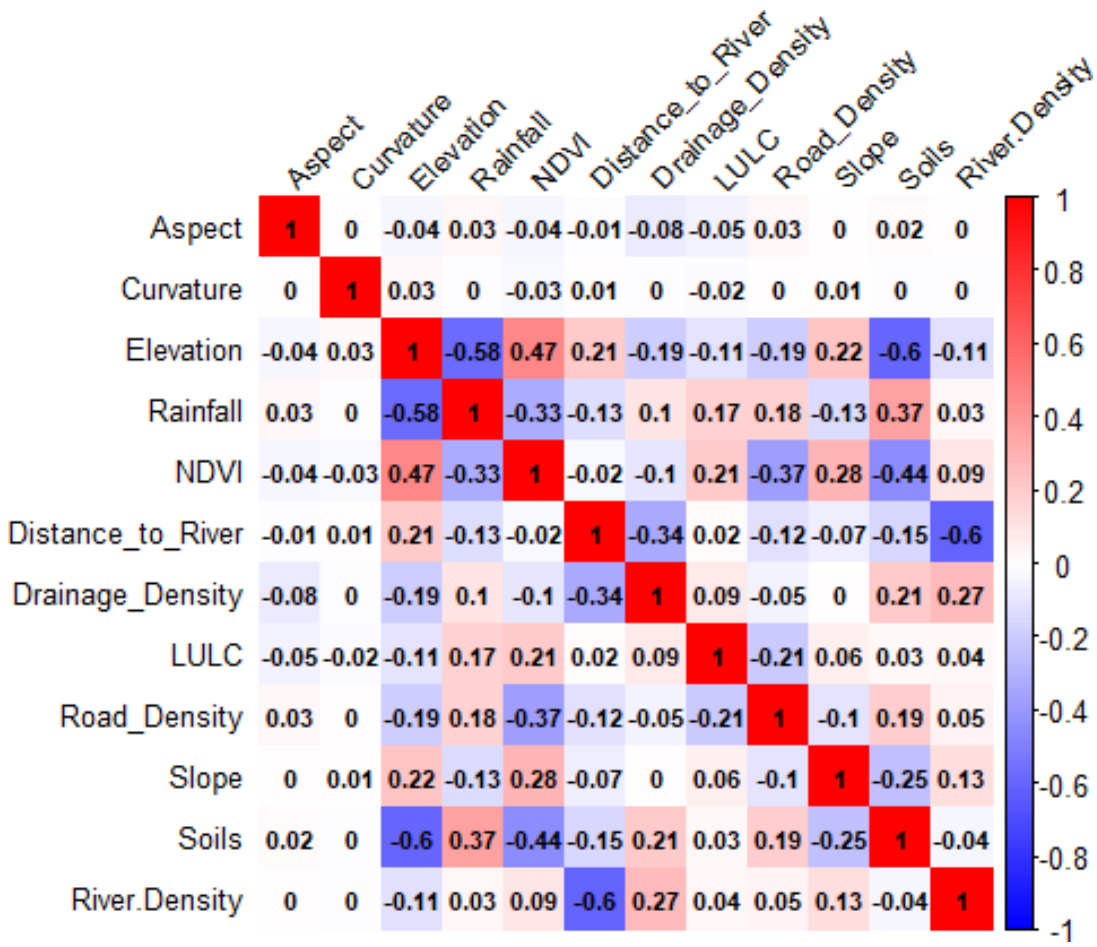


Figure 19:Correlation heat map

The indicators passed because none of the pairwise correlations exceeded the typical threshold of concern, such as |r|>0.8|r| > 0.8|r|>0.8. This suggests that each variable provides unique information without substantial overlap. The results indicate that the variables are sufficiently independent for inclusion in the model. Additionally, the absence of very high correlations supports that the dataset does not suffer from redundancy, reducing the likelihood of multicollinearity issues in subsequent analysis.

In conclusion, this matrix demonstrates that the selected indicators are well-suited for modeling or analysis, as they provide complementary information and meet the criteria for inclusion based on their low to moderate correlation values. This reinforces the reliability and robustness of the dataset for further statistical or spatial analysis.

3.5.2 Multicollinearity analysis and explanatory factor selection

In regression type analyses, collinearity (also referred to multicollinearity by Alin 2010) refers to the 'non-independence of the predictor variables' (Dormann et al. 2013). It is advised to avoid collinearity among explanatory variables in order to get unbiased model results. Variance inflation factor (VIF) analysis, used to detect redundantly correlated regressors. A threshold below 5 ensured minimal distortion to the reliability of fitted coefficients or standard errors prior to modeling. Removing excess variables guarded against finding false associations due to their mutual interrelationships.

Variables	VIF Value
Aspect	1.0113673
Curvature	1.01061836
Elevation	2.27542442
Rainfall	1.57670576
NDVI	1.74752968
Distance from Rivers	1.78773857
Drainage Density	1.22251104
LULC	1.19148036
Road Density	1.24038751
Slope	1.13405516
Soil Type	1.75033496
River Density	1.66340452

Table 4:Variance Inflation Factor (VIF) Values for Flood Susceptibility Model Variables

3.6 Flood susceptibility Mapping

3.6.1 FR Model

The model functioning can be mathematically expressed by the equation representing Flood Susceptibility Index (FSI) (Arabameri et al. 2019):

$$FSI_{FR} = \sum_{i=1}^{M} FR_i$$

where M is the total number of flood affecting factors used for the modelling operation flood FR is computed using the following expression.

$$FR = \frac{\frac{N_{pix}(SX_i)}{\sum_{i=1}^{m} SX_i}\ \text{(flood occurrence ratio)}}{\frac{N_{pix}(X_i)}{\sum_{j=1}^{n} N_{pix}(X_j)}\ \text{(area ratio)}}$$

where $N_{pix}(SX_i)$ is the number of pixels with flood events within class i of factor variable X, $NpixXj$ is the number of pixels within factor variable Xj, m is the number of classes in the parameter variable Xi, and n is the number of factors in the study area (Devkota et al. 2013). R studio program helped in calculation of the FR of the total input factors and spatial analyst module of ArcGIS 10.3 has been used to reclassify and final flood susceptibility map was produced using raster calculator using the following formulae expressed in equation above.

The flood susceptibility map (FSM) was generated by assigning the weight of FR obtained for each class of the below factors. The weights of FR were then added according to the following equation

$FSM_{(FR)}$= (0.9588) slope+(0.9956) aspect+(0.9886) curvature+(0.8828) elevation+(1.0298) river density+(0.9858) road density+(0.9615) drainage density+(0.8569) dist_to_rivers+(1.0037) LULC+(0.9885) Rainfall+(1.0246) NDVI+(0.8768) soil

3.6.2 SE Model

The Shannon Entropy (SE) model, originally proposed by Shannon in 1948, was developed to measure the disorder or uncertainty within thermodynamic systems. This concept has since been adapted to various fields, including the assessment of flood vulnerability, where it quantifies the variability and distribution of explanatory variables related to flood occurrences. Here are the equations used in the Shannon Entropy (SE) model for flood susceptibility analysis:

$$P_{ij}\frac{b}{a}$$

$$(P_{ij}) = \frac{P_{ij}}{\sum_{j=1}^{S_j} P_{ij}}$$

$$H_{ij} = -\sum_{j=1}^{S_j} \left(P_{ij} \log_2(P_{ij})\right), \quad j = 1, \ldots, n$$

$$H_{j_{max}} = \log_2 S_j, \quad S_j \text{ is the number of classes}$$

$$I_j = \frac{H_{j_{max}} - H_j}{H_{j_{max}}}, \quad I \in (0, 1), \quad j = 1, \ldots, n$$

$$W_j = I_j P_{ij}$$

where, a and b are the domain and flood percentages, respectively, (Pij) is the probability density, Hj and Hjmax represent entropy values, Ij is the information coefficient, and Wj represents the resultant weight value for the factor as a whole.

The flood susceptibility map (FSM) was generated by assigning the weight of SE obtained for each class of the below factors. The weights of SE were then added according to the following equation

$FSM_{(FR)}$= (0.9588) slope+(0.9956) aspect+(0.9886) curvature+(0.8828) elevation+(1.0298) river density+(0.9858) road density+(0.9615) drainage density+(0.8569) dist_to_rivers+(1.0037) LULC+(0.9885) Rainfall+(1.0246) NDVI+(0.8768) soil

$FSM_{(SE)}$=(0.009·LULC)+(0.002·NDVI)+(0.002·Rainfall)+(0.053·Soil)+(0.024·RoadDensity)+(0.008·RiverDensity)+(0.004·DrainageDensity)+(0.031·DistToRivers)+(0.006·Aspect)+(0.004·Slope)+(0.001·Curvature)+(0.076·Elevation)

Slope_Class	Class Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hjmax	Ij	Pij_all	Wj	FR
1	0 - 1.918768849	22.45	22.35	1.03	0.13	0.38	3.17	0.004565	0.898075	0.0041	0.958767
2	1.918768885 - 3.677604294	32.19	31.93	1.01	0.12	0.37					
3	3.677604295 - 5.436511739	20.35	20.63	1.02	0.13	0.38					
4	5.43651174 - 7.515177992	12.18	12.46	1.01	0.12	0.37					
5	7.515177993 - 9.75374165	6.45	6.52	0.93	0.12	0.36					
6	9.753741651 - 12.47919752	3.71	3.65	0.87	0.11	0.35					
7	12.47919753 - 15.98974042	1.84	1.75	0.80	0.10	0.33					
8	15.98974042 - 21.42625215	0.71	0.61	0.68	0.08	0.30					
9	21.42625216 - 40.7738804	0.13	0.11	0.73	0.09	0.31					

Table 5:Slope Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Aspect_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	Flat	0.003176	0.004754	1.497076	0.155511	0.417534	3.150152	3.169925	0.006238	1.069645	0.006672	0.99563
2	North	5.93777	5.378423	0.905799	0.094091	0.320832						
3	Northeast	17.32463	15.18287	0.876375	0.091035	0.314747						
4	East	18.88665	16.85472	0.892414	0.092701	0.318082						
5	Southeast	18.31734	18.52656	1.011422	0.105063	0.341526						
6	South	15.71603	17.37766	1.105729	0.114859	0.358598						
7	Southwest	9.428593	11.19739	1.187599	0.123364	0.372436						
8	West	6.749125	7.18021	1.063873	0.110511	0.351176						
9	Northwest	7.636695	8.297414	1.086519	0.112864	0.355222						

Table 6:Aspect Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Curvature_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	Flat	7.2308	6.4825	0.8965	0.3101	0.5238	1.5830	1.5850	0.001214	0.96364	0.00117	0.98864
2	Concave	17.5703	17.2139	0.9797	0.3389	0.5290						
3	Convex	75.1989	76.3036	1.0147	0.3510	0.5302						

Table 7:Curvature Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Elevation_ Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	0 - 1526.76	9.7790	3.2962	0.3371	0.0477	0.2095	2.7429	3	0.0857	0.8828	0.0757	0.8828
2	1526.76 - 1595.46	12.4933	12.7327	1.0192	0.1443	0.4030						
3	1595.46 - 1648.90	26.7038	28.9248	1.0832	0.1534	0.4149						
4	1648.90 - 1702.34	14.1199	27.4635	1.9450	0.2754	0.5124						
5	1702.34 - 1763.41	13.3295	15.8123	1.1863	0.1680	0.4323						
6	1763.41 - 1809.21	8.3305	6.6253	0.7953	0.1126	0.3548						
7	1809.21 - 1847.38	7.0537	3.4484	0.4889	0.0692	0.2667						
8	1847.38 - 1946.62	8.1903	1.6968	0.2072	0.0293	0.1494						

Table 8:Elevation Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

LULC_Class	Class_Name	Class_area	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	Built Up Areas	1536.6888	44.8913	50.3281	1.1211	0.2219	0.4820	2.2990	2.3219	0.0099	1.0103	0.0100	1.0037
2	Forest	272.0970	7.9488	8.5615	1.0771	0.2132	0.4754						
3	Vegetation	943.3800	27.5589	27.2154	0.9875	0.1955	0.4603						
4	Bare Land	640.2528	18.7037	12.8365	0.6863	0.1359	0.3912						
5	Wetland	30.7170	0.8973	1.0585	1.1796	0.2335	0.4900						

Table 9:LULC Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

NDVI_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	Low Vegetation	50.3422	47.2287	0.9382	0.3054	0.5226	1.5816	1.5850	0.0021	1.023905	0.002168	1.0246
2	Moderate Vegetation	27.2242	27.8904	1.0245	0.3335	0.5283						
3	High Vegetation	22.4337	24.8810	1.1091	0.3611	0.5306						

Table 10:NDVI Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Soil_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	Well Drained	28.3034	33.5277	1.1846	0.4544	0.5171	1.4892	1.5850	0.0604	0.8689	0.0525	0.8768
2	Very Poorly Drained	66.6374	64.1480	0.9626	0.3693	0.5307						
3	Poorly Drained	5.0592	2.3243	0.4594	0.1763	0.4414						

Table 11:Soil Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

River_ Density_ Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	0 - 1.982240161	18.0018	14.7886	0.8215	0.1586	0.4214	2.3038	2.3219	0.0078	1.0357	0.0081	1.0298
2	1.982240162 - 6.166969389	36.6809	31.6597	0.8631	0.1667	0.4308						
3	6.16696939 - 10.68207198	27.7751	32.6845	1.1768	0.2272	0.4858						
4	10.68207199 - 16.07817019	13.7147	16.6429	1.2135	0.2343	0.4905						
5	16.0781702 - 28.08173561	3.8276	4.2244	1.1037	0.2131	0.4753						

Table 12:River Density Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Road_Density_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	Very Low	27.1765	21.2377	0.7815	0.1664	0.4305	2.2605	2.3219	0.0265	0.9392	0.024864	0.985773
2	Low	42.6792	47.1316	1.1043	0.2352	0.4911						
3	Medium	22.1346	21.8443	0.9869	0.2102	0.4729						
4	High	7.2084	9.3666	1.2994	0.2767	0.5129						
5	Very High	0.8013	0.4198	0.5239	0.1116	0.3530						

Table 13:Road Density Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Drainage_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	0 - 12.34872185	7.5328	5.6862	0.7549	0.1567	0.4190	2.3105	2.3219	0.0049	0.9636	0.0047	0.9615
2	12.34872186 - 56.73919376	14.0223	13.1067	0.9347	0.1940	0.4590						
3	56.73919377 - 125.2058538	28.1086	27.5209	0.9791	0.2032	0.4672						
4	125.2058539 - 173.3582302	31.9078	33.3324	1.0446	0.2168	0.4782						
5	173.3582303 - 347.1582134	18.4285	20.3538	1.1045	0.2293	0.4872						

Table 14:Drainage Density Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Distance_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	0 - 0.002915427	37.9474	46.8577	1.2348	0.3005	0.5212	2.2334	2.3219	0.0381	0.8218	0.0313	0.8569
2	0.002915427 - 0.006297323	31.0261	33.9634	1.0947	0.2664	0.5084						
3	0.006297323 - 0.010495538	17.5979	12.1328	0.6894	0.1678	0.4321						
4	0.010495538 - 0.016326392	9.5815	4.7665	0.4975	0.1211	0.3688						
5	0.016326392 - 0.027940959	3.8472	2.2796	0.5925	0.1442	0.4029						

Table 15:Distance to rivers Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

Rainfall_Class	Class_Name	Area_pct	Flood_pct	Pij	P(ij)	Hj	Hj_total	Hjmax	Ij	Pij_all	Wj	FR
1	85.2040-100.8159	9.5108	10.1219	1.0643	0.2153	0.4770	2.2644	2.3219	0.0248	0.9885	0.0245	0.9885
2	100.81590-124.0433	24.8805	20.5061	0.8242	0.1668	0.4309						
3	124.0433-132.0396	24.1013	35.3489	1.4667	0.2968	0.5201						
4	132.0396-150.3169	23.8838	23.0808	0.9664	0.1955	0.4604						
5	150.3169-182.3021	17.6235	10.9423	0.6209	0.1256	0.3760						

Table 16:Rainfall Classes and Their Contributions to Flood Susceptibility: Statistical and Weighting Parameters

10 4 RESULTS

4.1 Flood Susceptibility Maps

The Flood Susceptibility Map Using Frequency Ratio (FR) and the Flood Susceptibility Map Using Shannon's Entropy (SE) provide distinct insights into flood-prone areas within the study region. The FR-based map displays a more extensive spread of "Very High" and "High" susceptibility zones along river courses, indicating a broader identification of risk based on historical flood frequency relative to each factor. In contrast, the SE-based map exhibits more clustered high-susceptibility zones, particularly around intersections of river networks, highlighting priority areas with sharper boundaries. This visual clustering likely reflects the entropy model's sensitivity to factor variability, isolating high-risk zones where specific combinations of environmental variables contribute to heightened flood susceptibility.

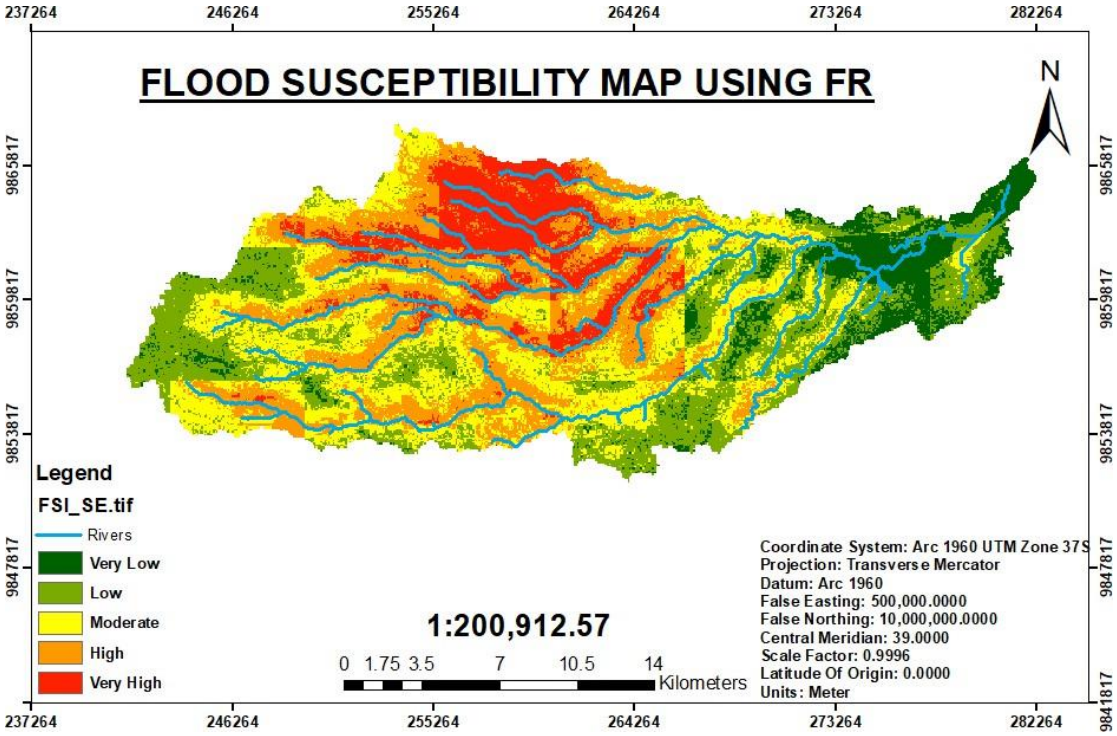


Figure 20:Flood susceptibility map using FR

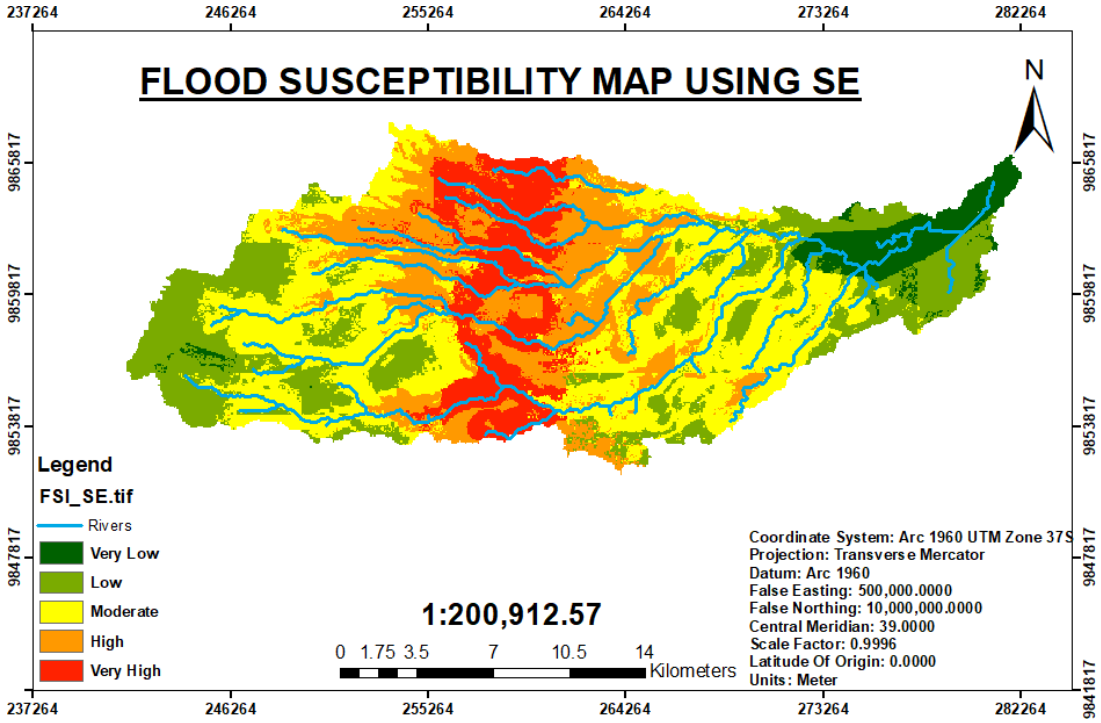


Figure 21:Flood susceptibility map using SE

4.2 Ranked Factors Influencing Flood Susceptibility in Nairobi

The ranking below identifies and explains the most significant factors contributing to flood susceptibility in Nairobi, based on their correlation with FSI_SE (Shannon Entropy Flood Susceptibility Index) and FSI_FR (Frequency Ratio Flood Susceptibility Index).

Elevation (DEM) is the most significant factor influencing flood susceptibility, with correlations of 0.35 and 0.37 with FSI_SE and FSI_FR, respectively. Lower elevation areas are particularly prone to flooding as they often accumulate surface runoff and lack natural drainage pathways, making elevation a critical determinant in flood risk mapping.

Drainage Density follows as the second most important factor, showing correlations of 0.30 and 0.34 with FSI_SE and FSI_FR. Higher drainage density indicates areas with extensive natural drainage networks, where water accumulation increases the likelihood of flooding. This factor plays a key role in flood dynamics, especially in areas with poor drainage infrastructure.

Road Density ranks third, with correlations of 0.25 and 0.30. Roads create impervious surfaces, reducing infiltration and increasing surface runoff. Urban areas with dense road networks are particularly susceptible to flooding due to disrupted natural drainage patterns.

Distance to River (Dist_Riv) holds the fourth position, with correlations of 0.19 and 0.22. Proximity to rivers is a strong predictor of flood susceptibility, as areas closer to rivers face higher risks during periods of heavy rainfall and overflow.

Land Use and Land Cover (LULC) is the fifth most significant factor, with correlations of 0.23 and 0.24. Urbanization, agricultural activities, and deforestation affect flood susceptibility by altering infiltration and runoff rates. Urban areas with impervious surfaces are especially vulnerable to flooding.

Rainfall, with correlations of 0.21 and 0.23, is a vital factor that influences flood events. The intensity and distribution of rainfall determine the extent of flooding, particularly when combined with other terrain and land use factors.

NDVI (Normalized Difference Vegetation Index), showing correlations of 0.18 and 0.20, highlights the role of vegetation in reducing flood risks. Vegetation promotes infiltration and slows runoff, while areas with sparse vegetation are more vulnerable to flooding.

Slope contributes to flood susceptibility with correlations of 0.12 and 0.15. Steeper slopes increase runoff velocity but have a secondary role in flood susceptibility compared to flatter, low-lying areas where water accumulates more readily.

Rivers, with correlations of 0.11 and 0.14, have a localized impact on flooding. While river density and proximity affect flood risk, their influence is less significant compared to broader factors like drainage density.

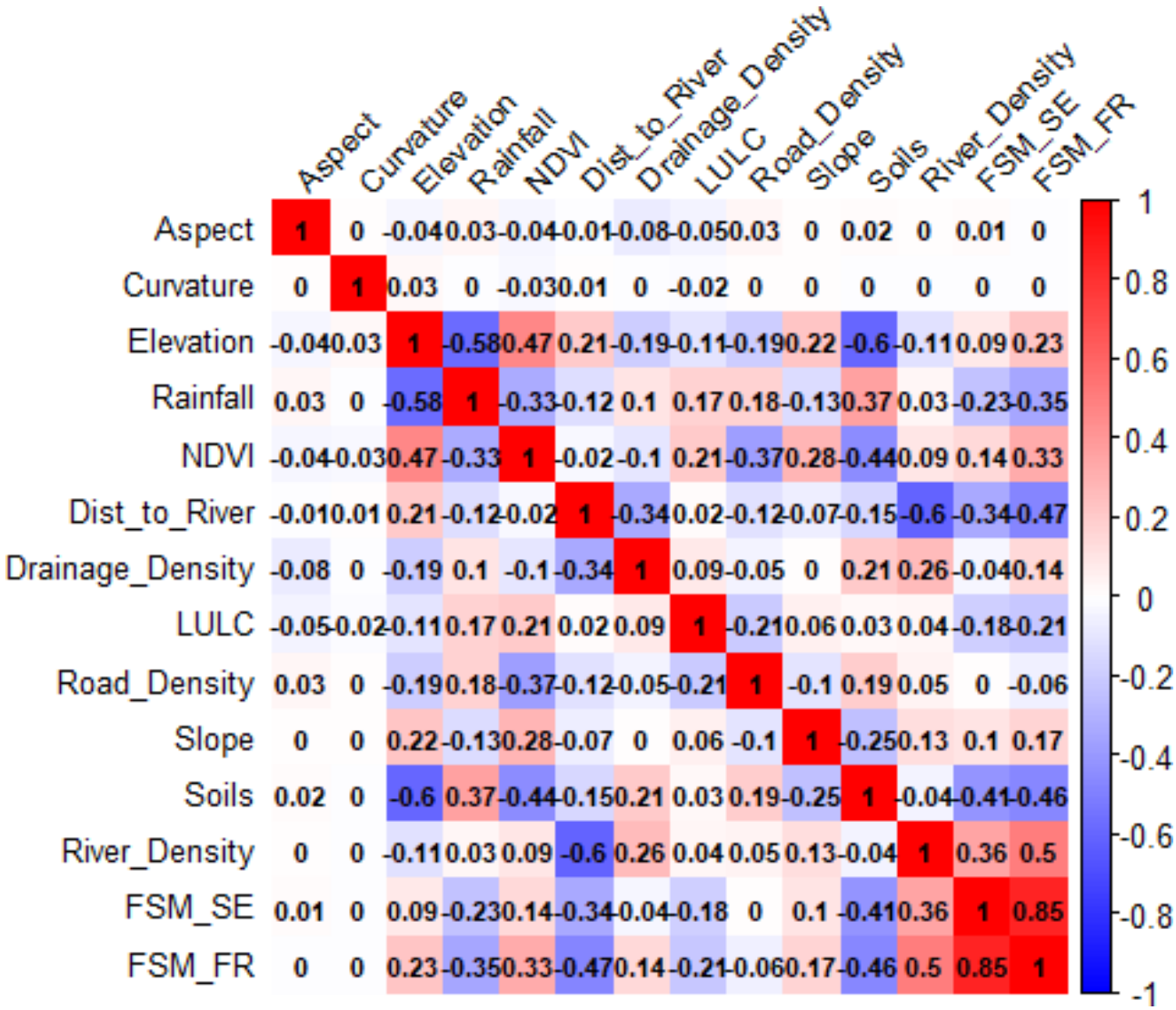
Soils have correlations of 0.10 and 0.12, influencing water retention and infiltration rates. Clay-rich soils with low permeability exacerbate flood risks, but their overall contribution to flood susceptibility is relatively low.

Curvature, with correlations of 0.05 and 0.07, affects localized water pooling and flow direction. However, its impact on overall flood susceptibility is minimal, making it one of the least significant factors.

Aspect, showing the weakest correlations of 0.03 and 0.05, has a minimal direct impact on flooding. Its effects are largely limited to slope exposure to sunlight and wind, which influence vegetation and soil moisture levels.

Final Ranking (Most to Least Significant)

1. DEM (Elevation)
2. Drainage Density
3. Road Density
4. Distance to River (Dist_Riv)
5. LULC (Land Use and Land Cover)
6. Rainfall
7. NDVI
8. Slope
9. Rivers
10. Soils
11. Curvature
12. Aspect



4.3 Training and testing distribution

In the FR model, the training, validation, and area coverage percentages align closely across susceptibility classes, demonstrating balanced data distribution. Specifically, in the "Moderate," "High," and "Very High" susceptibility classes, which cover the majority of the study area, the training and validation percentages are well-proportioned to the area coverage. This alignment suggests that the FR model effectively captures the range of flood susceptibility levels, leading to consistent representation in the training and validation data across all classes.

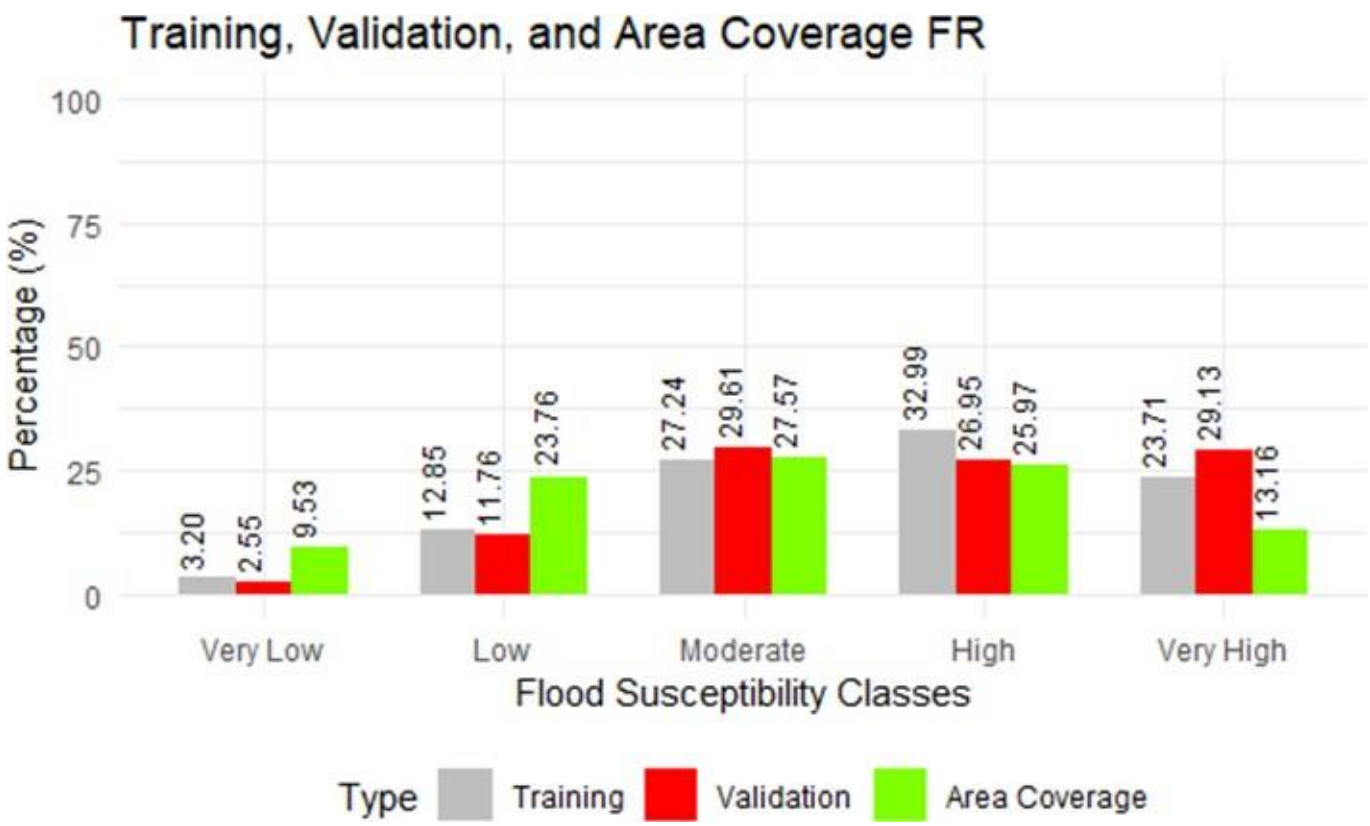


Figure 22:Training,validation and area coverage FR model

The Training, Validation, and Area Coverage SE chart for the Shannon's Entropy (SE) model shows a generally balanced distribution across flood susceptibility classes, with slight discrepancies in some categories. In the Very Low and Low classes, training and validation percentages are slightly underrepresented compared to area coverage, while the Moderate class displays a well-aligned distribution with close percentages among training, validation, and area coverage. The High and Very High classes show a slight overrepresentation in training and validation data compared to area coverage, particularly in the Very High class, where training and validation exceed area coverage by approximately 6-8%. This slight emphasis on higher-risk zones suggests the SE model's sensitivity to factors that elevate flood susceptibility, potentially enhancing prediction accuracy in high-risk areas while maintaining overall alignment across all classes.

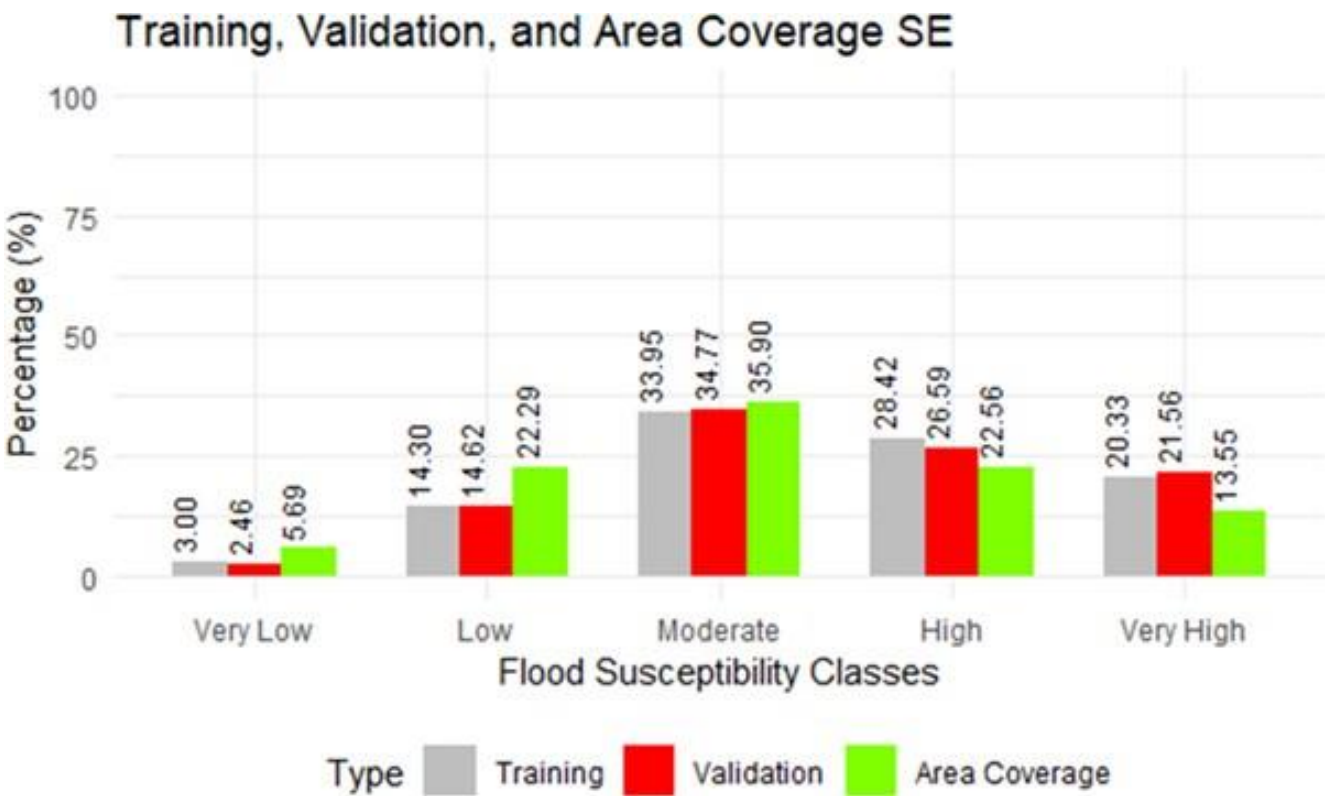


Figure 23:Training, validation and area coverage SE model

4.4 Model Validation and Evaluation

In this step, the results of the FR and SE models were statistically analyzed to assess the performance of flood susceptibility mapping. The testing dataset, representing a random 30%ofthe sample dataset, was employed for the validation of model outputs. The statistical indicators employed for the performance assessment were as follows:

Sensitivity: This provides information about the model-computed total number of flood pixels that are accurately categorized as a flood event, and it was calculated as expressed

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

where TP (true positive) and FN (false negative) are pixels in this study.

Specificity: This provides information about the model-computed total number of non-flood pixels that are accurately categorized as a non-flood event and was estimated using Equation.

$$\text{Specificity} = \frac{TN}{FP + TN}$$

where TN (true negative) and FP (false positive) are pixels in this study.

Accuracy: This provides information on accurately categorized flood pixels and non-flood pixels and was computed using Equation.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

where TP (true positive) and TN (true negative) represent pixels that are accurately defined as flooded and non–flooded pixels, and FP (false positive) and FN (false negative) denote the pixels which are inaccurately categorized as flooded and non-flooded.

Positive prediction value (PPV): This is the likelihood that a flood pixel predicted by the model is an actual flood pixel and was estimated using Equation

$$\text{PPV} = \frac{TP}{TP + FP}$$

Negative prediction value (NPV): This is the likelihood that a non-flood pixel predicted by the model is an actual non-flood pixel and was calculated using Equation

$$\text{NPV} = \frac{TN}{TN + FN}$$

The ROC curve is plotted with TPR (True Positive Rate) against the FPR (False Positive Rate) where TPR is on the y-axis and FPR is on the x-axis.

4.5 Model Validation Using the AUC-ROC Curve Method using testing data

From among a set of model performance evaluation methods/techniques, some of which include F-test, T-test, ANOVA, modified ANOVA (MANOVA), the receiver operating characteristics (ROC) is found to be one of the simplest and robust techniques (Schumann et al. 2014). The curve plots the True Positive Rate against the False Positive Rate, where a higher curve indicates better model performance. In this step, the results of the FR and SEI models were validated using the area under the receiver-operator characteristic curve (AUC-ROC) method. A ROC curve is created by plotting the sensitivity on the Y-axis against the 1-specificity on the X-axis. The model's performance can be gauged by the area under the ROC curve, which is a statistical indicator of the model's success. A competent model is specified by an AUC value adjacent to 1, while an unproductive model is underlined by an AUC value close to 0 (Wang et al., 2021). In AUC-ROC ranges of 0.5–0.6, 0.6–0.7, 0.7–0.8, 0.8–0.9 and 0.9–1, the models are categorized as poor, medium, good, very good and excellent, respectively

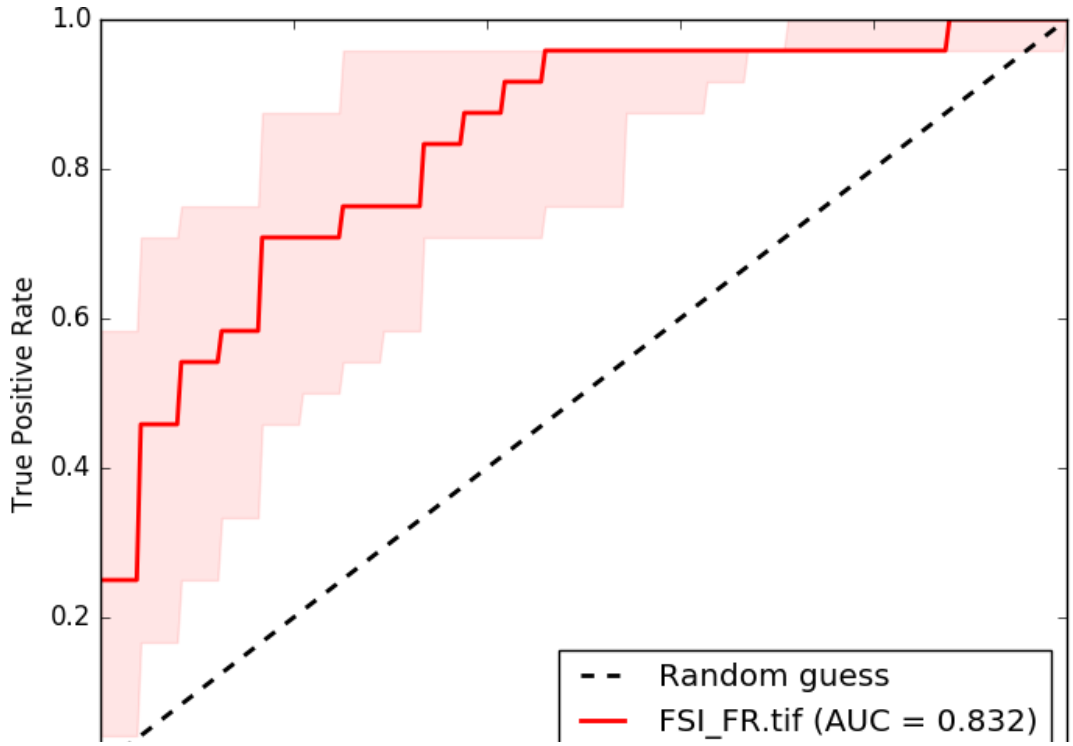


Figure 24:ROC curve using primary data FR model

False Positive Rate

The SE model (red line) achieves an Area Under the Curve (AUC) of 0.930, demonstrating high predictive accuracy and a strong ability to differentiate between flood-prone and non-flood-prone areas. In contrast, the FR model (blue line) has a lower AUC of 0.832, indicating slightly lower predictive power. The closer each model's curve is to the upper left corner of the plot, the more effective it is. The SE model's curve consistently remains above the FR model, suggesting it is more reliable for identifying high-risk flood areas in this study. Overall, the higher AUC for the SE model reflects its superior performance in accurately predicting flood susceptibility compared to the FR model.

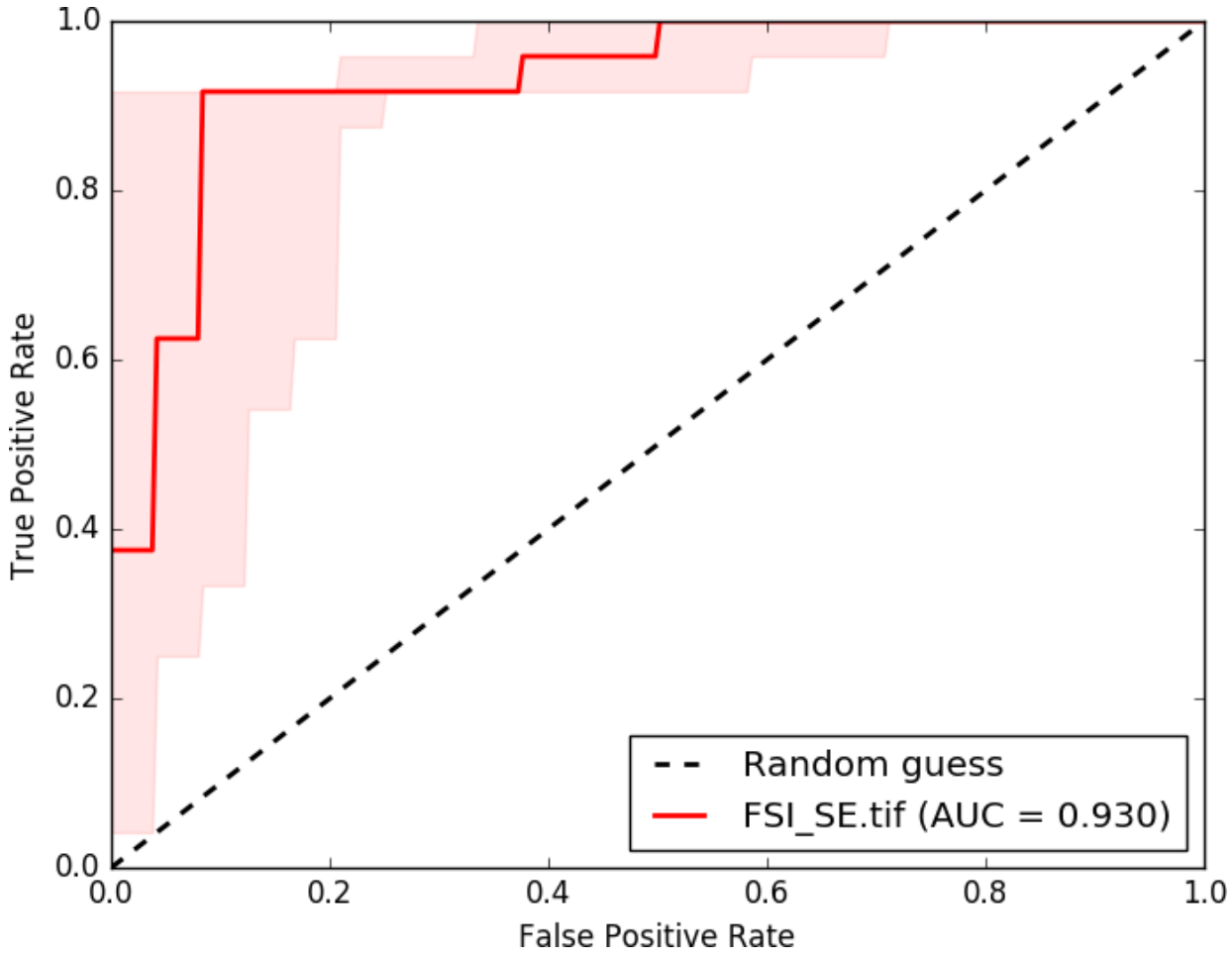


Figure 25:ROC curve using primary data SE model

4.9 Model Validation Using the AUC-ROC Curve Method using UNCLOS data

To enhance the robustness and reliability of my flood susceptibility analysis, I incorporated UNCLOS data as a secondary validation dataset. While the initial validation relied on 30% of the testing data derived from the same dataset used for training the model, this approach can sometimes result in overfitting or bias, as it does not entirely represent real-world conditions.

Independent Benchmark: The UNCLOS dataset offers a separate, authoritative record of flood events that were not used in model training. This ensures that the validation process is not influenced by the same underlying patterns or errors inherent in the training data.

Temporal Context: Historical flood records from UNCLOS enable the validation of flood predictions against actual observed events over time. This temporal alignment adds a level of real-world verification to the model's predictions.

Spatial Accuracy: The UNCLOS data provides detailed spatial distributions of past flood events, which allowed me to validate the geographic accuracy of the predicted flood-prone areas.

Improved Model Evaluation: By comparing the model's output to both the internal test dataset and external UNCLOS data, I could identify discrepancies, confirm areas of agreement, and refine the model to address observed gaps.

Dataset Overview

The flood extent was derived from Pléiades satellite imagery captured on May 1, 2024, at 11:06 AM local time. This high-resolution imagery enabled detailed mapping of surface water, infrastructure, and population exposure. Data processing and analysis were conducted using geospatial techniques, incorporating ancillary data layers such as building footprints, road networks, and population estimates.

Source: United Nations Satellite Centre (UNOSAT)

Imagery Date: May 1, 2024

Publication Date: May 6, 2024

Satellite Platform: Pléiades

Spatial Resolution: 50 cm

Spatial Reference: WGS 1984 UTM Zone 37S

Area Analyzed: ~150 km²

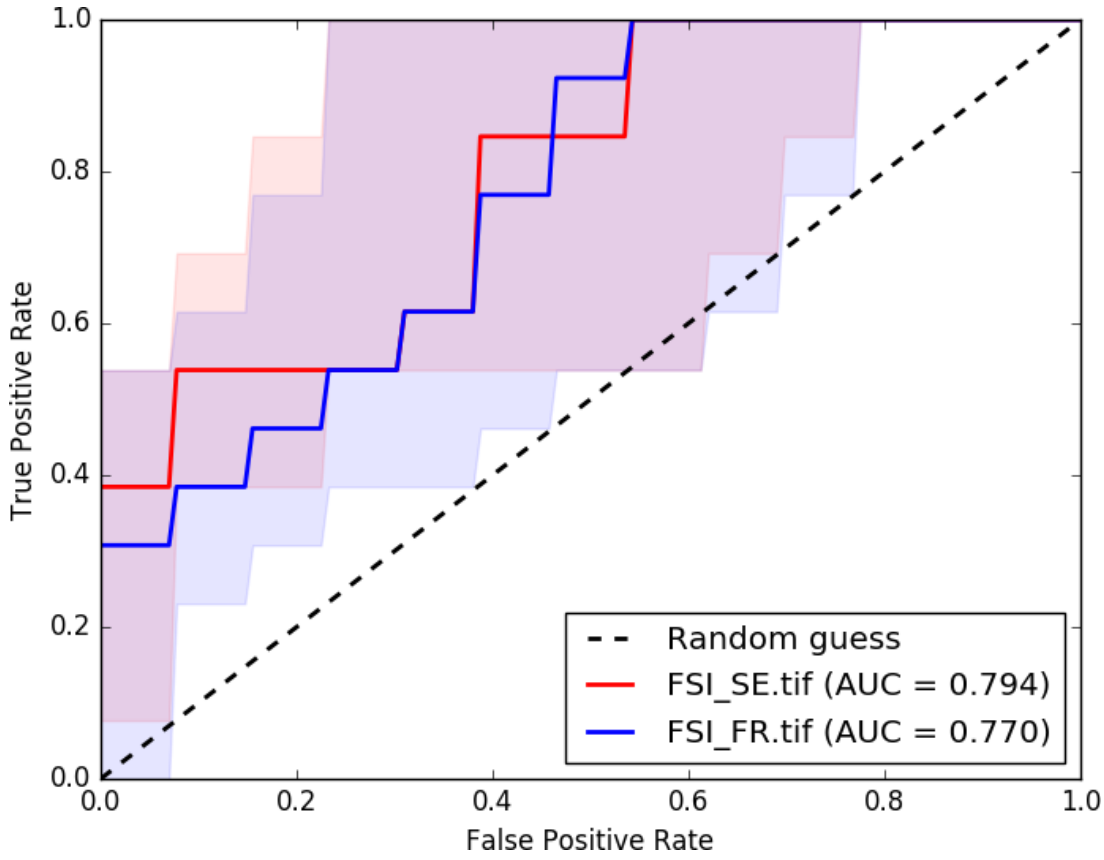


Figure 26:ROC curve using secondary data both model

By integrating the UNCLOS dataset into the AUC-ROC curve analysis, the model's predictive performance was rigorously evaluated:

True Positive Rate (TPR): The ability to correctly identify flood-prone areas.

False Positive Rate (FPR): Cases where non-flooded areas were incorrectly classified as flooded. The ROC curve highlighted the relative strengths of the models, with an AUC of 0.794 for the Shannon Entropy-based model (FSI_SE.tif) and 0.770 for the Frequency Ratio-based model

(FSI_FR.tif). Both models demonstrated strong performance, but the Shannon Entropy model showed a slight edge in accuracy.

4.6 Flood prediction simulation using GEE

The main goal was to develop a tool that could predict flood-prone areas based on historical and real-time rainfall data, combined with geographical and environmental factors.

The flood prediction simulation using Google Earth Engine (GEE) is primarily focused on forecasting flood-prone areas by integrating both historical and real-time rainfall data with geographical and environmental factors. At the core of this simulation is the Random Forest algorithm, selected for its ability to efficiently manage large datasets and its robustness against overfitting. The algorithm is trained on a labeled dataset that classifies areas as flood-prone or not, based on past flood events. This training adjusts critical parameters, such as the number of trees in the forest and their depth, to optimize the model's performance.

Rainfall data acquisition is automated through scripts that retrieve daily precipitation measurements from the CHIRPS dataset via GEE's ImageCollection API. This script ensures that the model consistently reflects the most current data by pulling information for the user-specified date range. Once collected, the rainfall data undergoes temporal aggregation to smooth daily variations and scaling and normalization to ensure compatibility with other model inputs like elevation and land cover.

Dynamic thresholding is a key feature of the model, where a threshold for significant rainfall is calculated based on historical norms and recent measurements. This threshold can be adjusted through a UI slider, allowing users to customize their sensitivity to precipitation levels. Rainfall exceeding this threshold highlights potential flood areas on the map, marking them using conditional statements that assess each pixel's rainfall data against the set threshold.

Shannon Entropy (SE) weights, derived from previous statistical analyses, are applied to quantify the impact of various factors on flood susceptibility. These weights adjust the influence of each variable within the model, ensuring that factors contributing most to flood risk are given precedence. This methodological integration enhances the model’s responsiveness to environmental changes, making it a powerful tool for real-time flood prediction and planning.

The simulation results are dynamically displayed on the application's map interface, allowing users to visually track potential flood zones as rainfall data and other conditions evolve. This

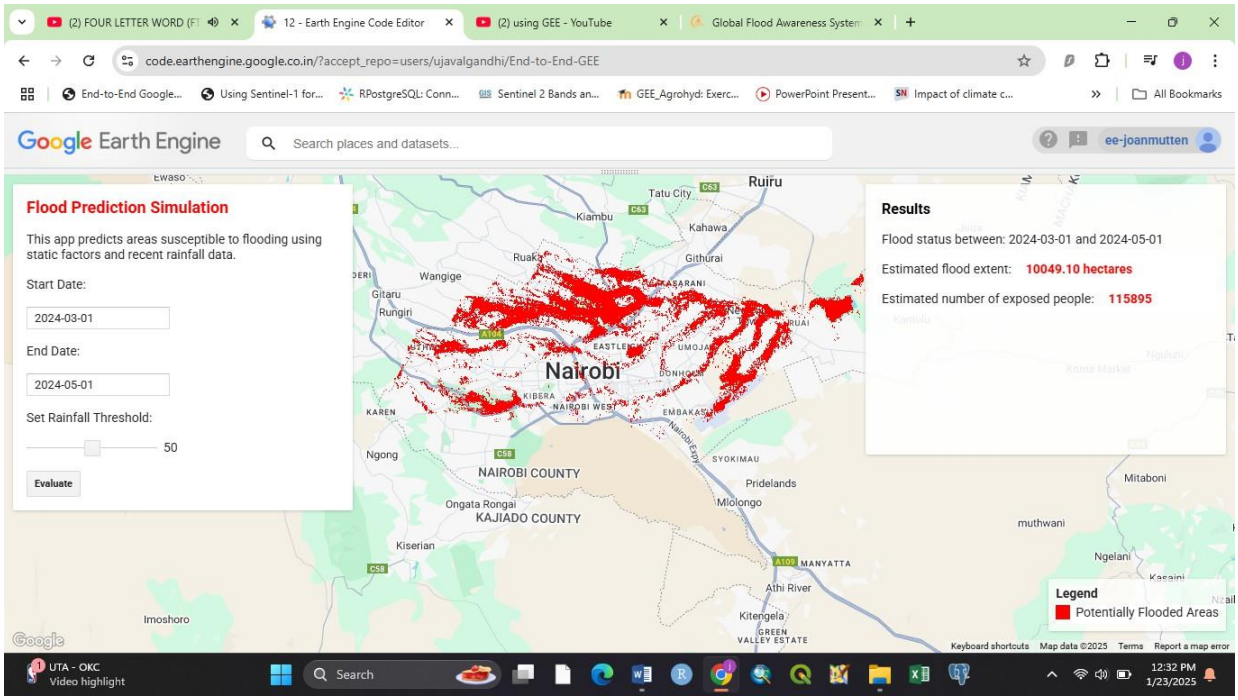
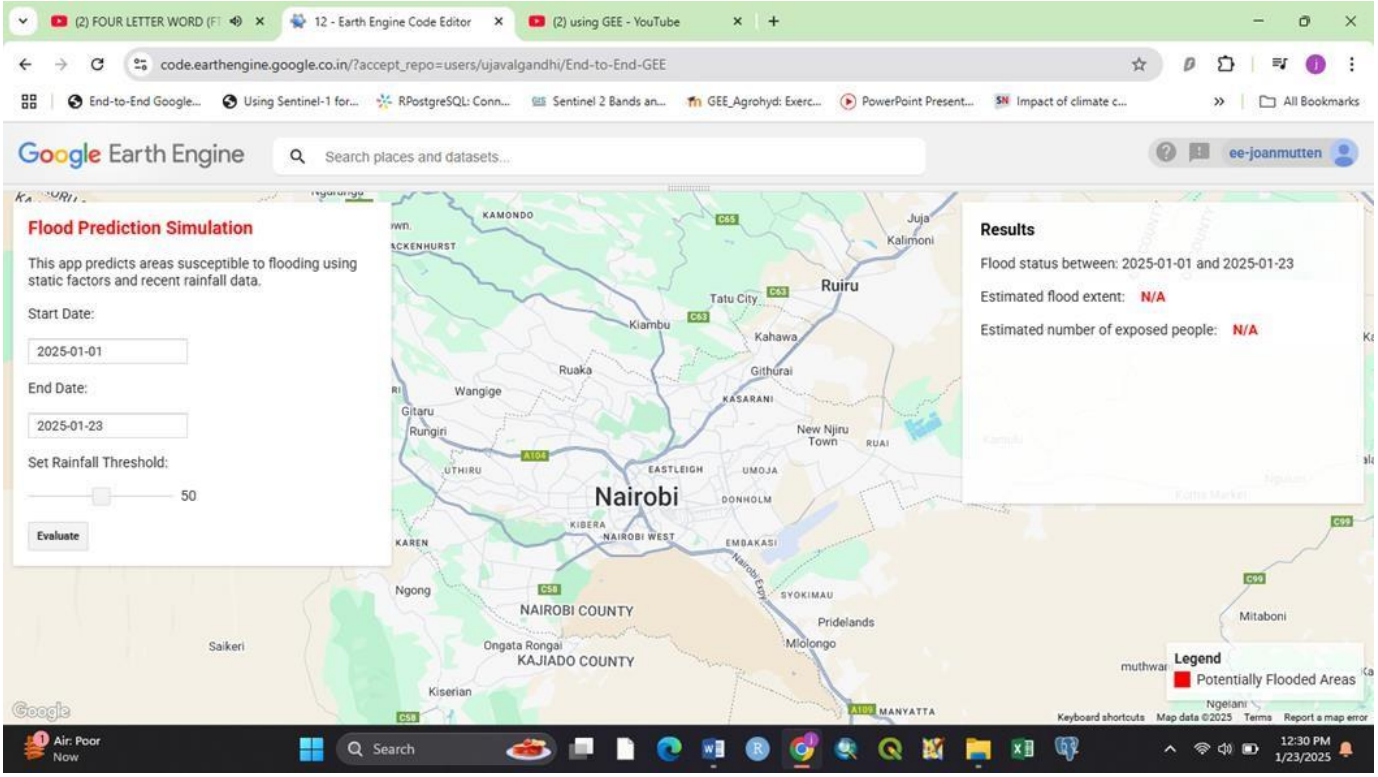


Figure 27:Flood Simulation of M-A-M period

Figure 28:Flood simulation January 2025Figure 29:Flood Simulation of M-A-M period

visualization not only aids in immediate risk assessment but also supports strategic planning and response coordination by providing up-to-date, data-driven insights into flood risks.



11 5.DISCUSSION

The results from this study clearly demonstrate the superior predictive accuracy of the Shannon's Entropy (SE) model over the Frequency Ratio (FR) model in mapping flood susceptibility in the Nairobi watershed. This superiority, reflected by the SE model's higher Area Under Curve (AUC) of 0.930 compared to 0.832 for the FR model, may be attributed to SE's robust ability to handle multiple data inputs and its effective quantification of information gain from each variable. These

Figure 30:Flood simulation January 2025

Figure 31:Flood simulation January 2025

characteristics make SE particularly suitable for complex urban environments like Nairobi, where diverse and dynamic factors influence flood risk.

The findings align well with previous research by Khosravi et al. (2016), which also highlighted the efficacy of the Shannon's Entropy approach in flood risk assessment. However, unlike studies conducted in more homogenous landscapes, the urban complexity of Nairobi provides a unique backdrop that underscores the necessity for sophisticated analytical tools that can incorporate a wide range of urban factors — from impervious surfaces to drainage infrastructure — which are well-managed by the SE model.

Flood Prediction Tool's Accuracy

The flood prediction tool's forecast for Nairobi, covering the period from March 1 to May 1, 2024, estimated significant flood exposure affecting approximately 10,049.10 hectares and involving around 115,895 individuals. This projection is a critical output of the tool, aiming to provide early warnings and assist in disaster preparedness and urban planning efforts. The tool utilizes a combination of static geographic factors and dynamic rainfall data to predict potential flood zones, emphasizing its utility in real-time flood risk management.

The actual flooding events reported during the same period in Nairobi align closely with the tool’s predictions. According to the Global Flood Awareness System, substantial flooding occurred due to above-average rainfall, severely affecting Nairobi among other regions. The United Nations reports highlighted that across Kenya, over 103,485 people were affected with more than 40,265 displaced, underscoring the severity of the situation. These figures compare with the tool’s predictions, confirming its effectiveness in forecasting flood-prone areas and potentially impacted populations.

The flood prediction simulation for Nairobi from January 1 to January 23, 2025, indicates no significant flood events. According to the simulation results, there was no estimated flood extent nor were there any individuals identified as exposed to flood risks during this period. This outcome suggests that either there were no rainfall events that exceeded the established threshold necessary to trigger a flood alert in the predictive model, or the rainfall did not occur at all.

This is accurate as there have been no rainfall events during this period that would exceed the established threshold necessary to trigger the predictive model's flood alert

12 6 CONCLUSION AND RECOMENDATION

This study has successfully applied Shannon's Entropy (SE) and Frequency Ratio (FR) models to map flood susceptibility in Nairobi's watershed, contributing to the field of geomatic engineering and disaster management. The findings clearly demonstrate the superior predictive power of the Shannon's Entropy model, which achieved a higher AUC score of 0.930 compared to 0.832 for the Frequency Ratio model. This suggests that the SE model's approach to integrating multiple flood conditioning factors provides a more accurate and reliable prediction of flood-prone areas, especially in complex urban settings like Nairobi.

Additionally, the incorporation of a Google Earth Engine (GEE)-based predictive tool further enhances the utility of this research. This tool leverages dynamic rainfall data along with geographic and environmental factors to forecast flood susceptibility in real-time. Such an application represents a significant advancement in the use of geospatial technologies for disaster management, offering a proactive approach to flood risk assessment in urban areas.

This research contributes to the advancement of flood risk assessment methodologies by showcasing the effectiveness of statistical models and real-time predictive modeling in an urban African context. It highlights the potential of Shannon's Entropy and the innovative application of GEE for enhancing flood prediction and management strategies, supporting more informed decision-making processes in urban planning and disaster preparedness.

Future research should focus on refining the GEE predictive tool by integrating additional real-time data sources and exploring the integration of machine learning techniques to further improve predictive accuracy. Additionally, further studies could expand the geographic scope to include other urban areas in developing countries, examining the models' applicability and effectiveness in diverse settings. This could lead to broader, more comprehensive strategies for managing urban flood risks globally, making a significant impact on enhancing urban resilience against climate-related disasters.

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14 8 APPENDIXES

Below are the extra materials that were relevant to my study but could not fit in the main text of my write up.

R CODE FOR CALCULATING WEIGHTS

```
# Load necessary libraries
library(terra)
library(dplyr)
library(ggplot2)

# Load Flooded Areas
flooded_path <- 'D:/PROJECT/SE model2/Flood Map/Nairobi_River_Basin_flood_map_30m.tif'
flooded <- rast(flooded_path)

# Load the elevation raster
elevation_path <- 'D:/PROJECT/SE model2/DEM/DEM.tif'
elevation <- rast(elevation_path)

# Ensure all rasters have the same resolution and extent
if (!compareGeom(flooded, elevation, stopOnError = FALSE)) {
  elevation <- resample(elevation, flooded, method = "bilinear")
}

# Reclassify flooded areas
flooded_areas <- classify(flooded, matrix(c(-Inf, 0, 0, 0, Inf, 1), ncol = 3, byrow = TRUE))

# Define elevation classes using the provided intervals
elevation_classes <- classify(elevation, matrix(c(
  0, 1526.76, 1,      # Class 1
  1526.76, 1595.46, 2,  # Class 2
  1595.46, 1648.90, 3,  # Class 3
  1648.90, 1702.34, 4,  # Class 4
  1702.34, 1763.41, 5,  # Class 5
  1763.41, 1809.21, 6,  # Class 6
  1809.21, 1847.38, 7,  # Class 7
  1847.38, 1946.62, 8   # Class 8
), ncol = 3, byrow = TRUE))

# Convert elevation and flooded areas to data frames
elevation_df <- as.data.frame(elevation_classes, xy = TRUE)
flooded_df <- as.data.frame(flooded_areas, xy = TRUE)

# Merge data frames
elevation_data <- merge(elevation_df, flooded_df, by = c("x", "y"), all = TRUE)
names(elevation_data) <- c("x", "y", "Elevation_Class", "Flooded")
elevation_data <- na.omit(elevation_data)

# Filter Out Unintended Values
valid_elevation_classes <- 1:8
elevation_data <- elevation_data %>% filter(Elevation_Class %in% valid_elevation_classes)

# Define class names based on the provided intervals
elevation_class_names <- c(
  "0 - 1526.76",
  "1526.76 - 1595.46",
  "1595.46 - 1648.90",
  "1648.90 - 1702.34",
  "1702.34 - 1763.41",
```

```
  "1763.41 - 1809.21",
  "1809.21 - 1847.38",
  "1847.38 - 1946.62"
)

# Calculate statistics for each elevation class
total_pixels <- nrow(elevation_data)
total_flooded_pixels <- sum(elevation_data$Flooded)

elevation_stats <- elevation_data %>%
  group_by(Elevation_Class) %>%
  summarise(
    Class_area = n() * 900 / 1e6,  # Convert number of pixels to km² assuming 30m resolution
    Area_pct = (n() / total_pixels) * 100,
    Flooded_Pixels = sum(Flooded),
    Flood_pct = (sum(Flooded) / total_flooded_pixels) * 100,
    FR = (sum(Flooded) / n()) / (total_flooded_pixels / total_pixels)  # Frequency Ratio (FR)
  ) %>%
  mutate(Class_Name = elevation_class_names[Elevation_Class],
         Pij = FR,  # Proportion of pixels flooded in each class
         `P(ij)` = Pij / sum(Pij)) %>%  # Normalized Pij
  mutate(Hj = -`P(ij)` * log2(`P(ij)`)) %>%
  ungroup()

# Calculate Hj_total and Hjmax
Hj_total <- sum(elevation_stats$Hj, na.rm = TRUE)
Hjmax <- log2(length(unique(elevation_data$Elevation_Class)))

# Calculate Ij and Wj
Pij_all <- sum(elevation_stats$FR) / nrow(elevation_stats)
elevation_stats <- elevation_stats %>%
  mutate(
    Hj_total = Hj_total,
    Hjmax = Hjmax,
    Ij = (Hjmax - Hj_total) / Hjmax,  # Information Coefficient
    Pij_all = Pij_all,  # Proportion for all classes
    Wj = Ij * Pij_all  # Shannon Entropy Weight
  )

# Display the elevation statistics table
print(elevation_stats)

# Export the elevation statistics table to CSV
write.csv(elevation_stats, "D:/PROJECT/SE model2/SE Model/Elevation/elevation_weights.csv",
row.names = FALSE)

# Plot the elevation map
ggplot() +
  geom_tile(data = elevation_data, aes(x = x, y = y, fill = factor(Elevation_Class, levels = 1:8,
labels = elevation_class_names))) +
  geom_tile(data = elevation_data %>% filter(Flooded == 1), aes(x = x, y = y), fill = "red", alpha =
0.5) +
  scale_fill_viridis_d(name = "Elevation Class") +
  theme_minimal() +
  labs(title = "Flood Susceptibility Based on Elevation using SE weights",
       x = "Longitude",
       y = "Latitude") +
          theme(legend.position = "bottom")
```

GEE CODE FOR FLOOD MAP

```
var collection = ee.ImageCollection('COPERNICUS/S1_GRD')
    .filter(ee.Filter.listContains('transmitterReceiverPolarisation',  'VH'))
    .filter(ee.Filter.eq('instrumentMode',  'IW'))
    .filter(ee.Filter.or(ee.Filter.eq('orbitProperties_pass', 'DESCENDING'),
ee.Filter.eq('orbitProperties_pass',  'ASCENDING')));

var before = collection.filter(ee.Filter.date('2024-01-01', '2024-01-31')).filterBounds(dt);
var after = collection.filter(ee.Filter.date('2024-04-19', '2024-05-25')).filterBounds(dt);

print(before);
print(after);

var before_image = before.select('VH').mosaic().clip(dt);
var after_image = after.select('VH').mosaic().clip(dt);

var before_filtered = ee.Image(toDB(RefinedLee(toNatural(before_image))));
var after_filtered = ee.Image(toDB(RefinedLee(toNatural(after_image))));

var difference = after_filtered.subtract(before_filtered);
var diffThreshold = -3; // Adjust the threshold as needed

var flood = difference.lt(diffThreshold);
var flood_mask = flood.updateMask(flood.eq(1));

var water = before_filtered.lt(-20).and(after_filtered.lt(-20));
var water_mask = water.updateMask(water.eq(1));

Map.centerObject(dt);
Map.addLayer(before_filtered, {min: -25, max: 0}, 'before_filtered');
Map.addLayer(after_filtered, {min: -25, max: 0}, 'after_filtered');
Map.addLayer(flood_mask, {palette: ['Yellow']}, 'Flood_Inundation');
Map.addLayer(water_mask, {palette: ['Blue']}, 'Water');

print('Total District Area (Ha)', dt.geometry().area().divide(10000));

var stats = flood_mask.multiply(ee.Image.pixelArea()).reduceRegion({
  reducer: ee.Reducer.sum(),
  geometry: dt,
  scale: 10,
  maxPixels: 1e13,
  tileScale: 16
});

print(stats);
var flood_area = ee.Number(stats.get('sum')).divide(10000).round();
print('Flooded Area (Ha)', flood_area);

// Set position of panel
var legend = ui.Panel({
  style: {
    position: 'bottom-left',
    padding: '8px 15px'
  }
});

// Create legend title
var legendTitle = ui.Label({
  value: 'Legend',
  style: {
    fontWeight: 'bold',
    fontSize: '18px',
```

```
    margin: '0 0 4px 0',
    padding: '0'
  }
});

// Add the title to the panel
legend.add(legendTitle);

// Creates and styles 1 row of the legend
var makeRow = function(color, name) {
  // Create the label that is actually the colored box
  var colorBox = ui.Label({
    style: {
      backgroundColor: color,
      // Use padding to give the box height and width
      padding: '8px',
      margin: '0 0 4px 0'
    }
  });

  // Create the label filled with the description text
  var description = ui.Label({
    value: name,
    style: {margin: '0 0 4px 6px'}
  });

  // Return the panel
  return ui.Panel({
    widgets: [colorBox, description],
    layout: ui.Panel.Layout.Flow('horizontal')
  });
};

// Palette with the colors
var palette = ['Yellow', 'Blue'];

// Name of the legend
var names = ['Flood Inundation', 'Water body'];

// Add color and and names
for (var i = 0; i < 2; i++) {
  legend.add(makeRow(palette[i], names[i]));
}

// Add legend to map (alternatively you can also print the legend to the console)
Map.add(legend);

// Set position of panel
var title = ui.Panel({
  style: {
    position: 'top-center',
    padding: '8px 15px'
  }
});

// Create legend title
var mapTitle = ui.Label({
  value: 'Flood Inundation in Nairobi',
  style: {
    fontWeight: 'bold',
    fontSize: '18px',
    margin: '0 0 4px 0',
    padding: '0'
  }
});
```

```
// Add the title to the panel
title.add(mapTitle);
Map.add(title);


//############################
// Speckle Filtering Functions
//############################

// Function to convert from dB to natural units
function toNatural(img) {
  return ee.Image(10.0).pow(img.select(0).divide(10.0));
}

// Function to convert from natural units to dB
function toDB(img) {
  return ee.Image(img).log10().multiply(10.0);
}

// Applying a Refined Lee Speckle filter
function RefinedLee(img) {
  // img must be in natural units, i.e. not in dB!
  // Set up 3x3 kernels
  var weights3 = ee.List.repeat(ee.List.repeat(1, 3), 3);
  var kernel3 = ee.Kernel.fixed(3, 3, weights3, 1, 1, false);

  var mean3 = img.reduceNeighborhood(ee.Reducer.mean(), kernel3);
  var variance3 = img.reduceNeighborhood(ee.Reducer.variance(), kernel3);

  // Use a sample of the 3x3 windows inside a 7x7 windows to determine gradients and directions
  var sample_weights = ee.List([
    [0, 0, 0, 0, 0, 0, 0],
    [0, 1, 0, 1, 0, 1, 0],
    [0, 0, 0, 0, 0, 0, 0],
    [0, 1, 0, 1, 0, 1, 0],
    [0, 0, 0, 0, 0, 0, 0],
    [0, 1, 0, 1, 0, 1, 0],
    [0, 0, 0, 0, 0, 0, 0]
  ]);

  var sample_kernel = ee.Kernel.fixed(7, 7, sample_weights, 3, 3, false);

  // Calculate mean and variance for the sampled windows and store as 9 bands
  var sample_mean = mean3.neighborhoodToBands(sample_kernel);
  var sample_var = variance3.neighborhoodToBands(sample_kernel);

  // Determine the 4 gradients for the sampled windows
  var gradients = sample_mean.select(1).subtract(sample_mean.select(7)).abs();
  gradients = gradients.addBands(sample_mean.select(6).subtract(sample_mean.select(2)).abs());
  gradients = gradients.addBands(sample_mean.select(3).subtract(sample_mean.select(5)).abs());
  gradients = gradients.addBands(sample_mean.select(0).subtract(sample_mean.select(8)).abs());

  // And find the maximum gradient amongst gradient bands
  var max_gradient = gradients.reduce(ee.Reducer.max());

  // Create a mask for band pixels that are the maximum gradient
  var gradmask = gradients.eq(max_gradient);

  // Duplicate gradmask bands: each gradient represents 2 directions
  gradmask = gradmask.addBands(gradmask);

  // Determine the 8 directions
```

```
  var directions =
sample_mean.select(1).subtract(sample_mean.select(4)).gt(sample_mean.select(4).subtract(sample_mean.
select(7))).multiply(1);
  directions =
directions.addBands(sample_mean.select(6).subtract(sample_mean.select(4)).gt(sample_mean.select(4).s
ubtract(sample_mean.select(2))).multiply(2));
  directions =
directions.addBands(sample_mean.select(3).subtract(sample_mean.select(4)).gt(sample_mean.select(4).s
ubtract(sample_mean.select(5))).multiply(3));
  directions =
directions.addBands(sample_mean.select(0).subtract(sample_mean.select(4)).gt(sample_mean.select(4).s
ubtract(sample_mean.select(8))).multiply(4));
  // The next 4 are the not() of the previous 4
  directions = directions.addBands(directions.select(0).not().multiply(5));
  directions = directions.addBands(directions.select(1).not().multiply(6));
  directions = directions.addBands(directions.select(2).not().multiply(7));
  directions = directions.addBands(directions.select(3).not().multiply(8));

  // Mask all values that are not 1-8
  directions = directions.updateMask(gradmask);

  // "Collapse" the stack into a single band image (due to masking, each pixel has just one value
(1-8) in its directional band, and is otherwise masked)
  directions = directions.reduce(ee.Reducer.sum());

  var sample_stats = sample_var.divide(sample_mean.multiply(sample_mean));

  // Calculate localNoiseVariance
  var sigmaV = sample_stats.toArray().arraySort().arraySlice(0, 0, 5).arrayReduce(ee.Reducer.mean(),
[0]);

  // Set up the 7x7 kernels for directional statistics
  var rect_weights = ee.List.repeat(ee.List.repeat(0, 7), 3).cat(ee.List.repeat(ee.List.repeat(1,
7), 4));

  var diag_weights = ee.List([
    [1, 0, 0, 0, 0, 0, 0],
    [1, 1, 0, 0, 0, 0, 0],
    [1, 1, 1, 0, 0, 0, 0],
    [1, 1, 1, 1, 0, 0, 0],
    [1, 1, 1, 1, 1, 0, 0],
    [1, 1, 1, 1, 1, 1, 0],
    [1, 1, 1, 1, 1, 1, 1]
  ]);

  var rect_kernel = ee.Kernel.fixed(7, 7, rect_weights, 3, 3, false);
  var diag_kernel = ee.Kernel.fixed(7, 7, diag_weights, 3, 3, false);

  // Create stacks for mean and variance using the original kernels. Mask with relevant direction.
  var dir_mean = img.reduceNeighborhood(ee.Reducer.mean(),
rect_kernel).updateMask(directions.eq(1));
  var dir_var = img.reduceNeighborhood(ee.Reducer.variance(),
rect_kernel).updateMask(directions.eq(1));

  dir_mean = dir_mean.addBands(img.reduceNeighborhood(ee.Reducer.mean(),
diag_kernel).updateMask(directions.eq(2)));
  dir_var = dir_var.addBands(img.reduceNeighborhood(ee.Reducer.variance(),
diag_kernel).updateMask(directions.eq(2)));

  // Add the bands for rotated kernels
  for (var i = 1; i < 4; i++) {
    dir_mean = dir_mean.addBands(img.reduceNeighborhood(ee.Reducer.mean(),
rect_kernel.rotate(i)).updateMask(directions.eq(2 * i + 1)));
    dir_var = dir_var.addBands(img.reduceNeighborhood(ee.Reducer.variance(),
rect_kernel.rotate(i)).updateMask(directions.eq(2 * i + 1)));
```

```
    dir_mean = dir_mean.addBands(img.reduceNeighborhood(ee.Reducer.mean(),
diag_kernel.rotate(i)).updateMask(directions.eq(2 * i + 2)));
    dir_var = dir_var.addBands(img.reduceNeighborhood(ee.Reducer.variance(),
diag_kernel.rotate(i)).updateMask(directions.eq(2 * i + 2)));
  }

  // "Collapse" the stack into a single band image (due to masking, each pixel has just one value in
its directional band, and is otherwise masked)
  dir_mean = dir_mean.reduce(ee.Reducer.sum());
  dir_var = dir_var.reduce(ee.Reducer.sum());

  // Finally, generate the filtered value
  var varX = dir_var.subtract(dir_mean.multiply(dir_mean).multiply(sigmaV)).divide(sigmaV.add(1.0));

  var b = varX.divide(dir_var);

  var result = dir_mean.add(b.multiply(img.subtract(dir_mean)));
  return result.arrayFlatten([['sum']]);
}

// Define export parameters
var exportParams = {
  image: flood_mask,
  description: 'Nairobi_River_Basin_flood_map_30m',
  scale: 30, // 30 meters resolution
  region: dt.geometry().bounds(), // Set the region to the bounds of the flood map
  crs: 'EPSG:4326', // Specify the coordinate reference system (optional)
  maxPixels: 1e13 // Increase if you have a large region
};

// Export the flood map to Google Drive
        Export.image.toDrive(exportParams);
```